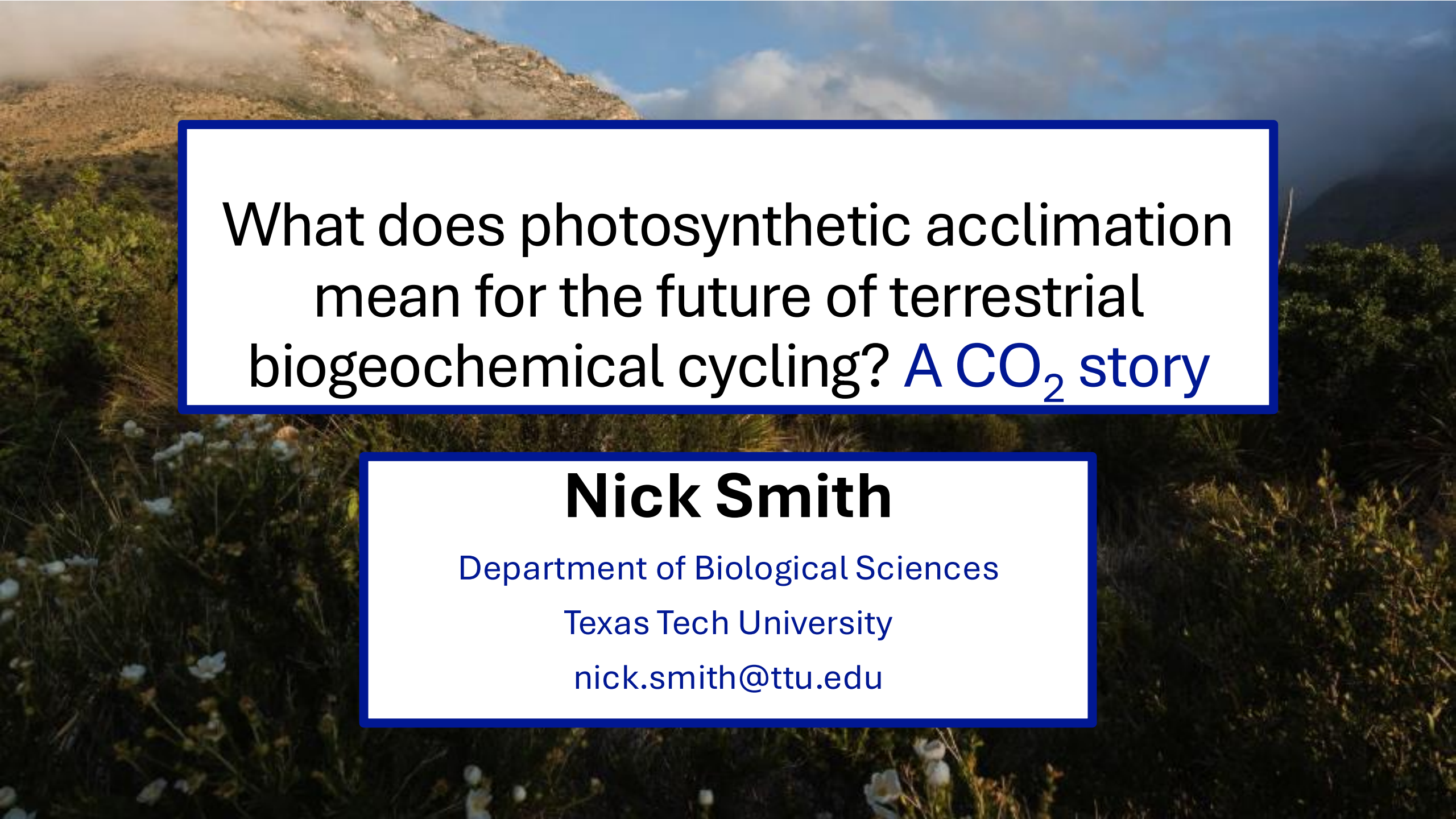


What does photosynthetic acclimation mean for the future of terrestrial biogeochemical cycling? A CO_2 story

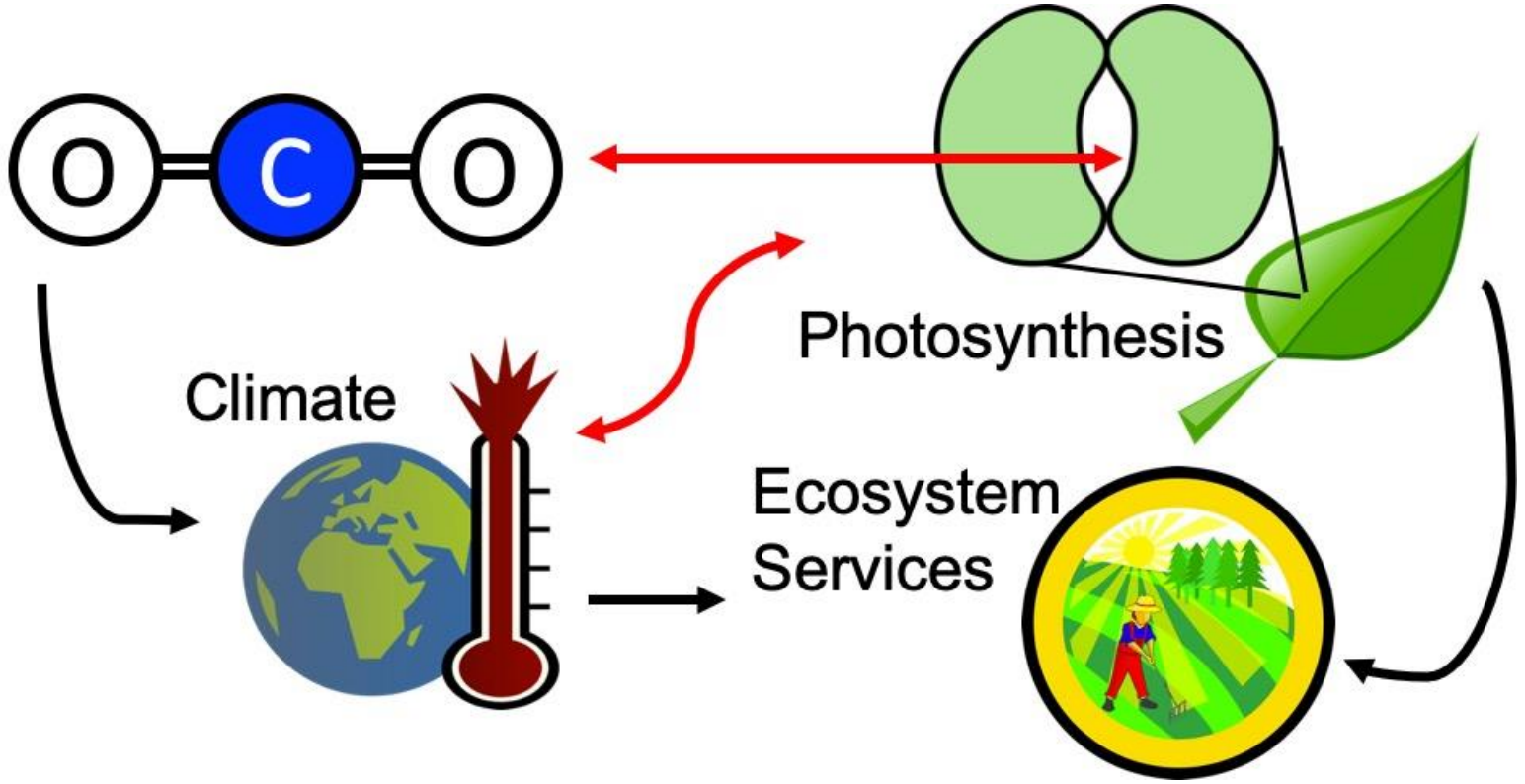
Nick Smith

Department of Biological Sciences

Texas Tech University

nick.smith@ttu.edu

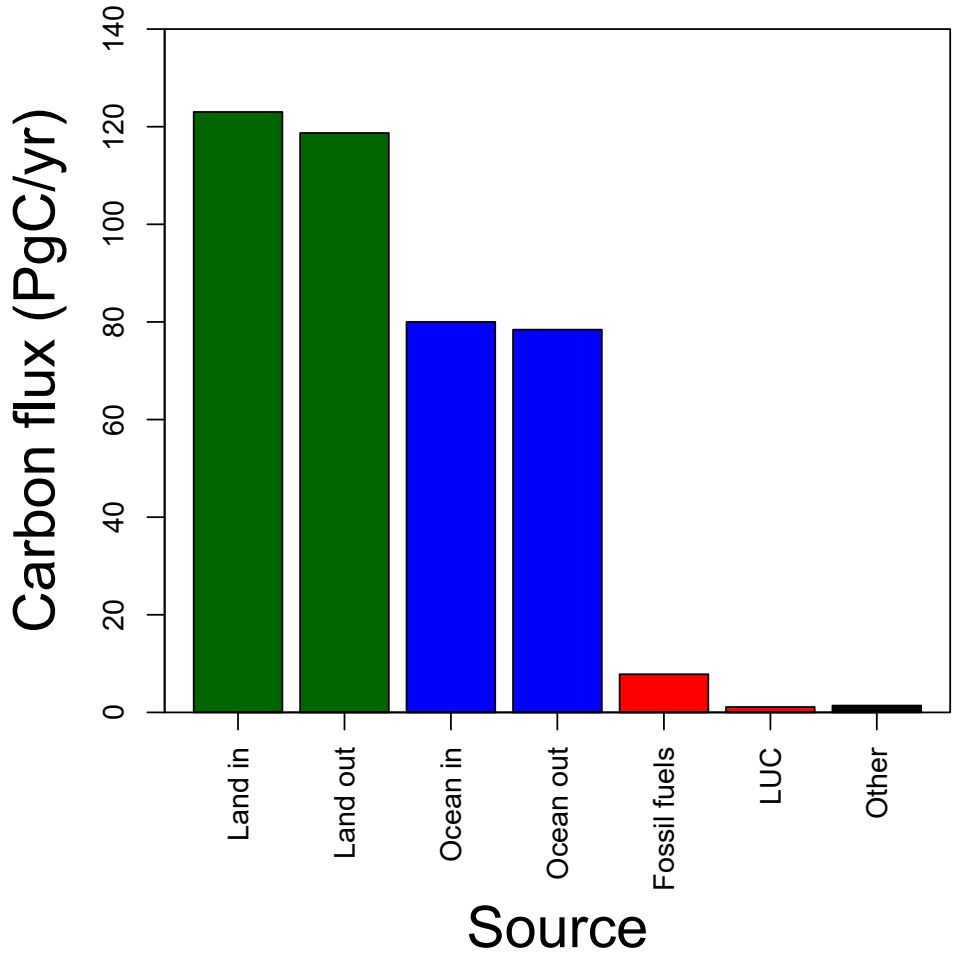
Our lab examines photosynthesis as a regulator of global change impacts on ecosystem services



Smith (2024) *Am J Botany*

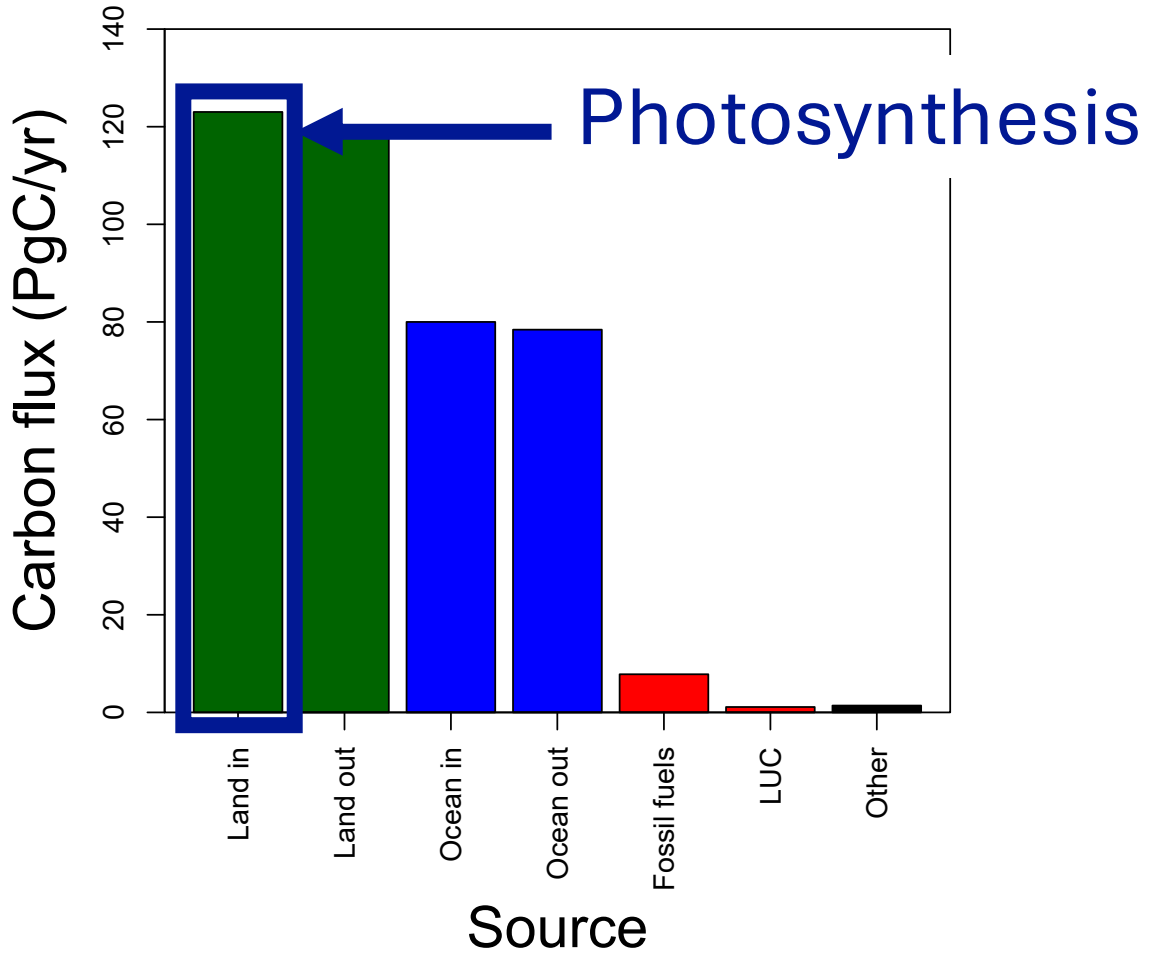
Photosynthesis is important! An example…

Photosynthesis is important! An example...



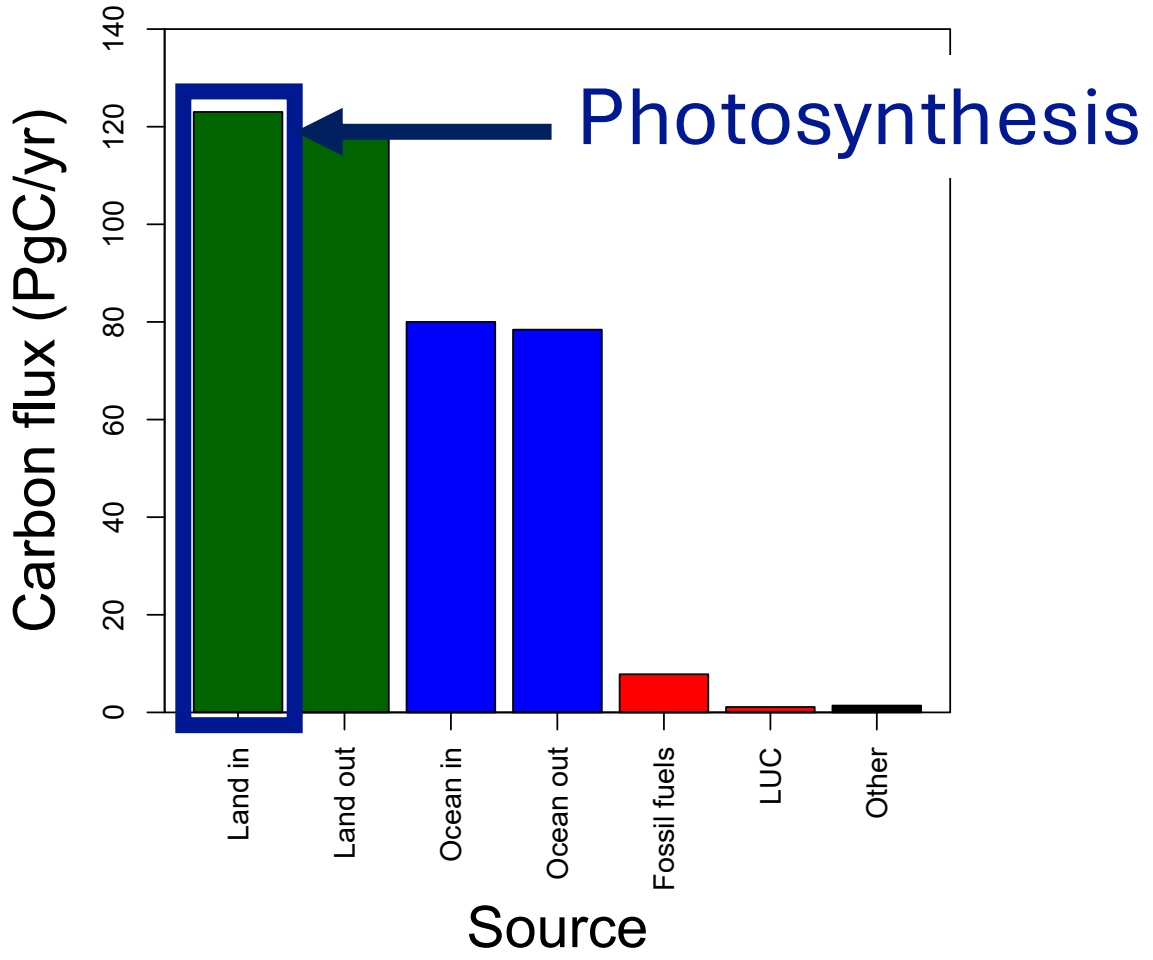
Data from IPCC (2013)

Photosynthesis is important! An example...



Data from IPCC (2013)

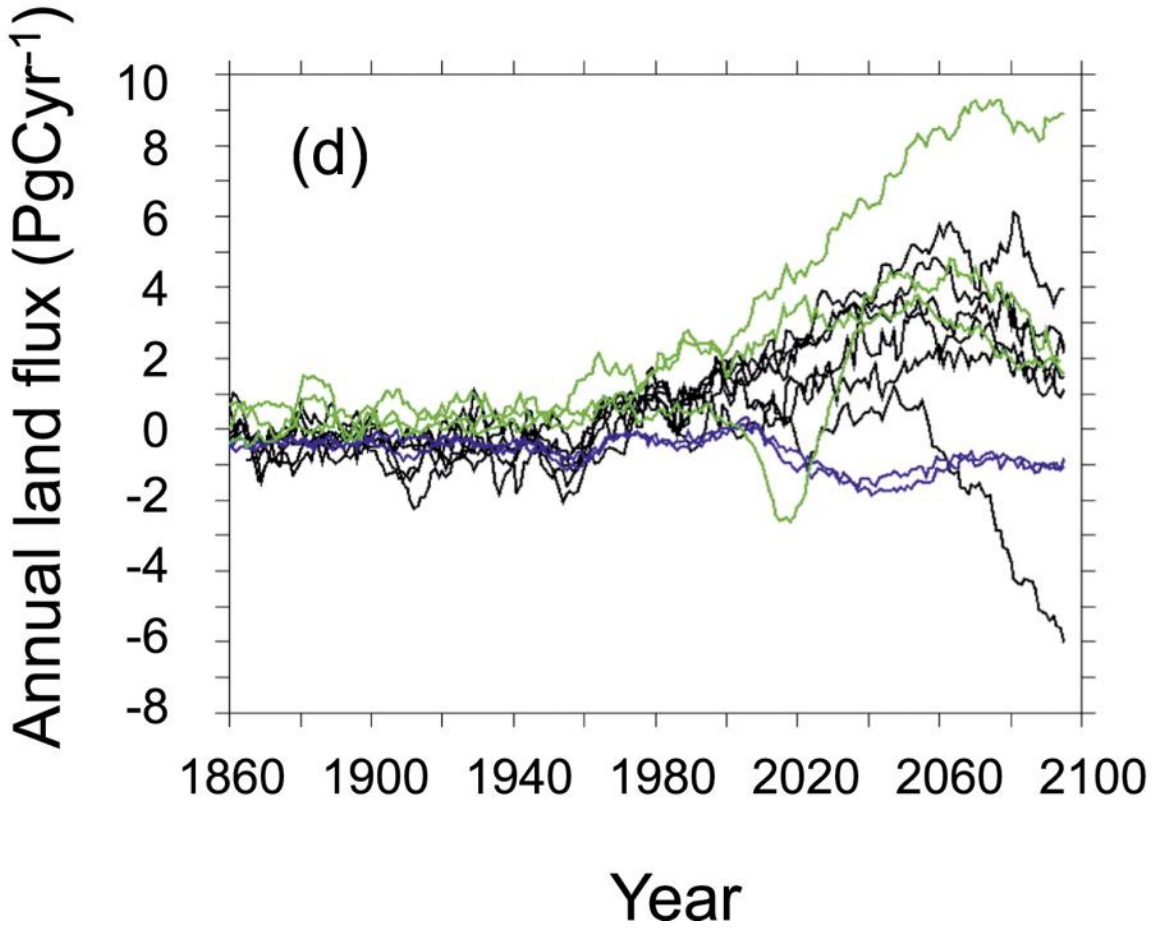
Photosynthesis is important! An example…



A small percentage change in photosynthesis can have large consequences for climate

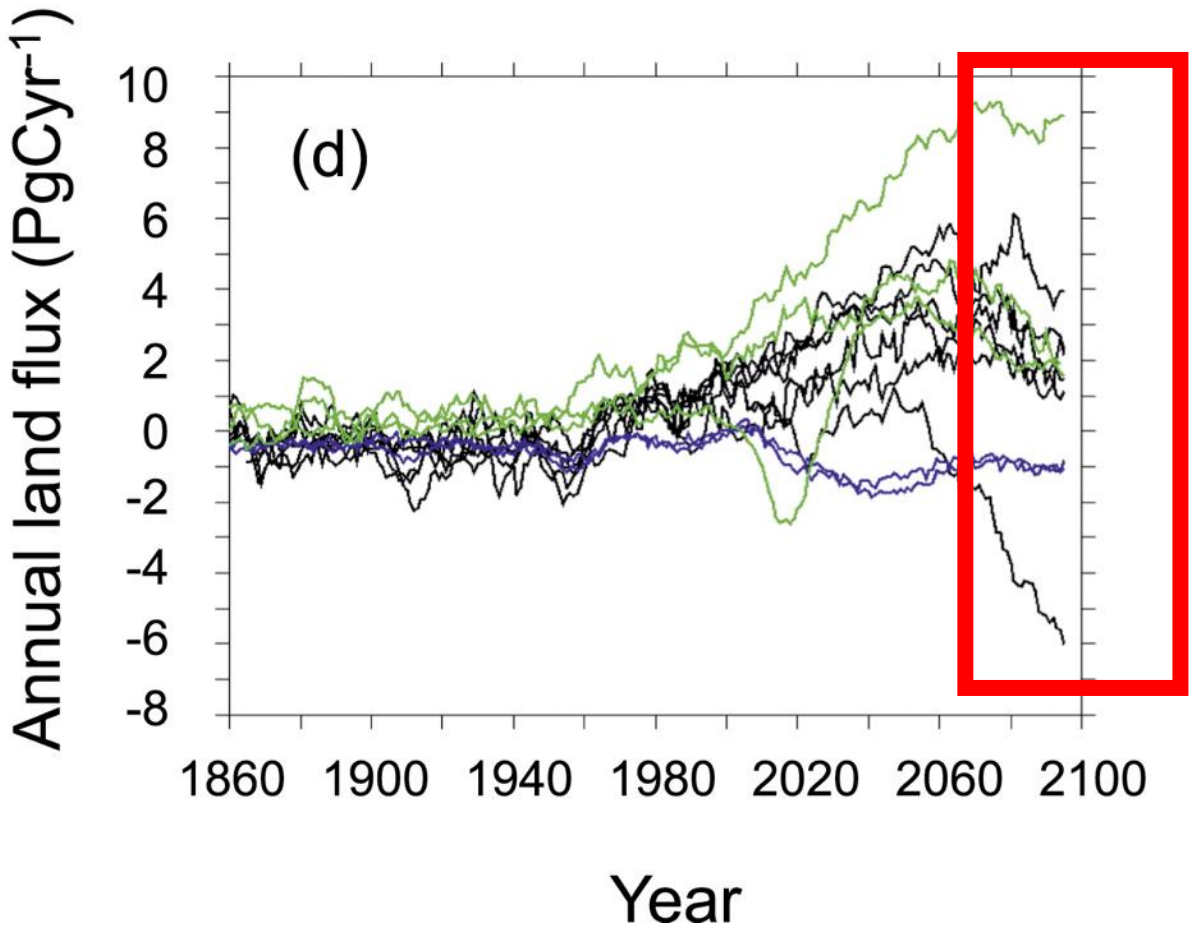
Data from IPCC (2013)

But predictions are uncertain!



Friedlingstein et al. (2014) *J Climate*

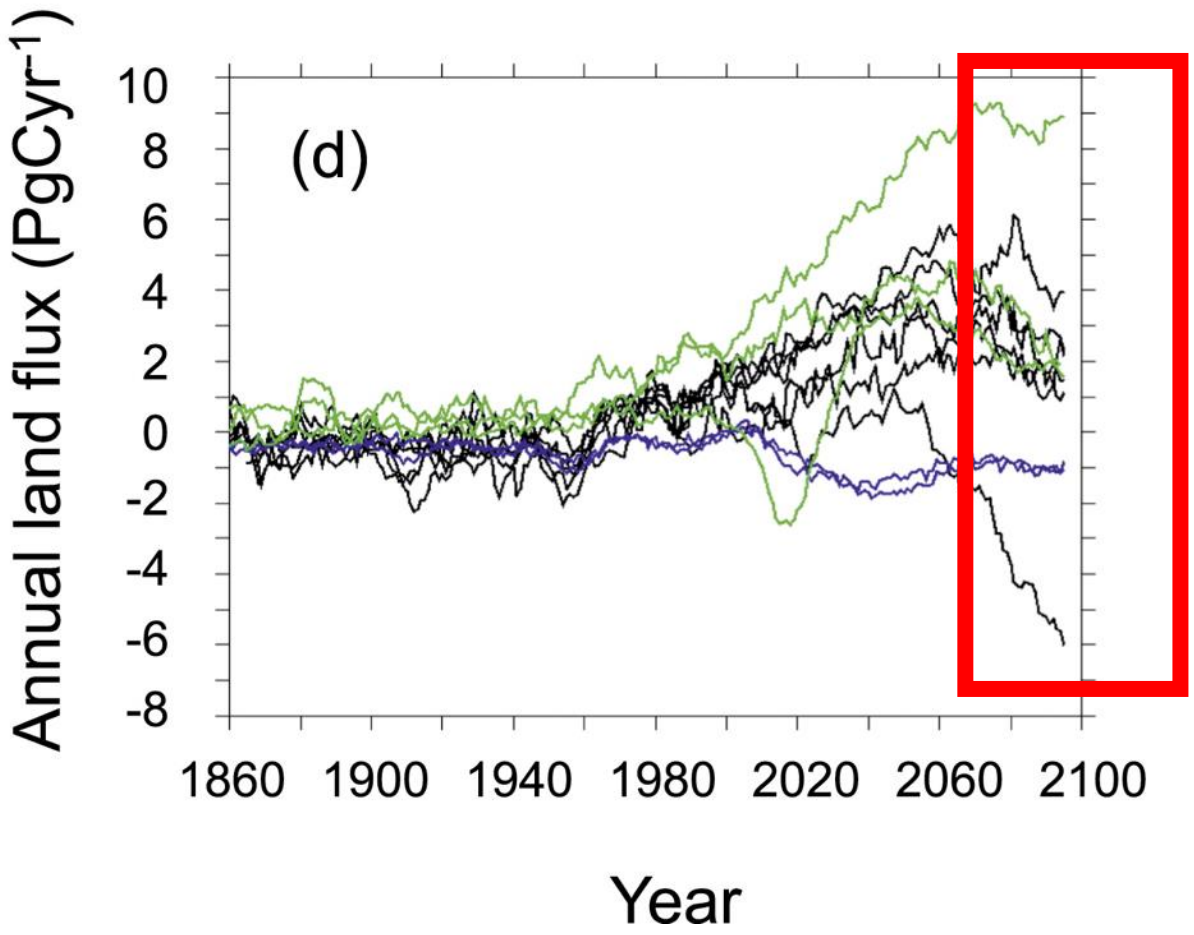
But predictions are uncertain!

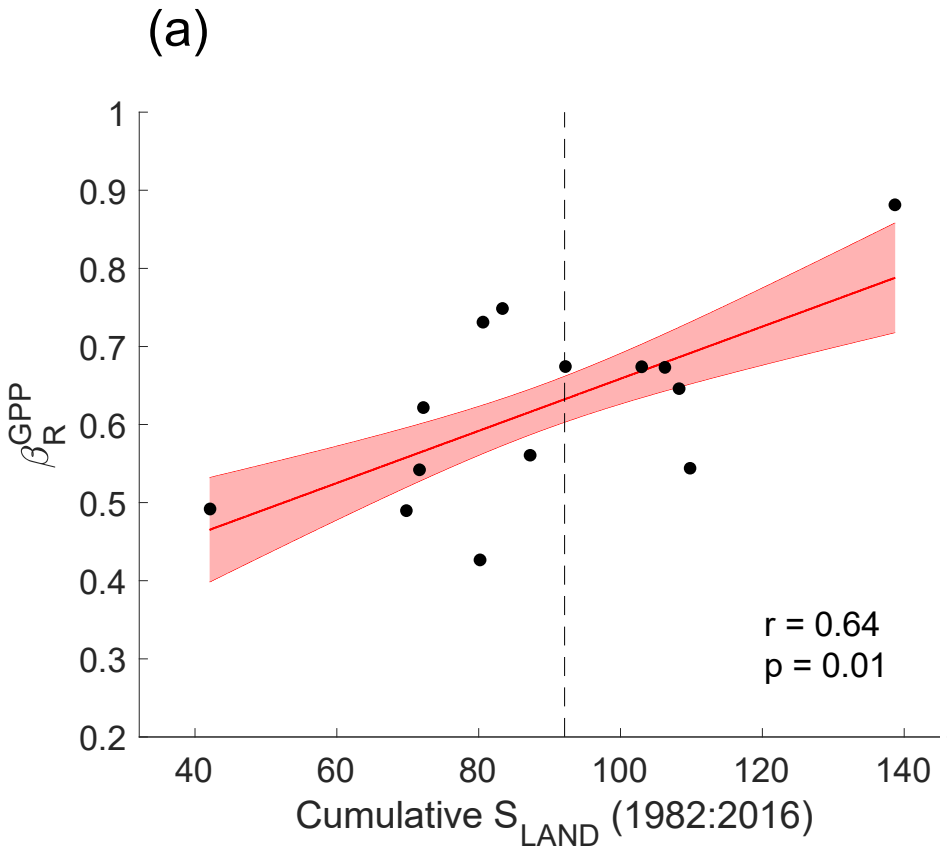


Future model uncertainty (14 Pg) > current fossil fuel emissions (9.5 Pg)

Friedlingstein et al. (2014) *J Climate*

This uncertainty is driven by uncertainty in photosynthesis





Friedlingstein et al. (2014) *J Climate*

Keenan et al. (2023) *Nature Climate Change*

Why the uncertainty? Theoretical models for photosynthesis exist

Planta 149, 78–90 (1980)

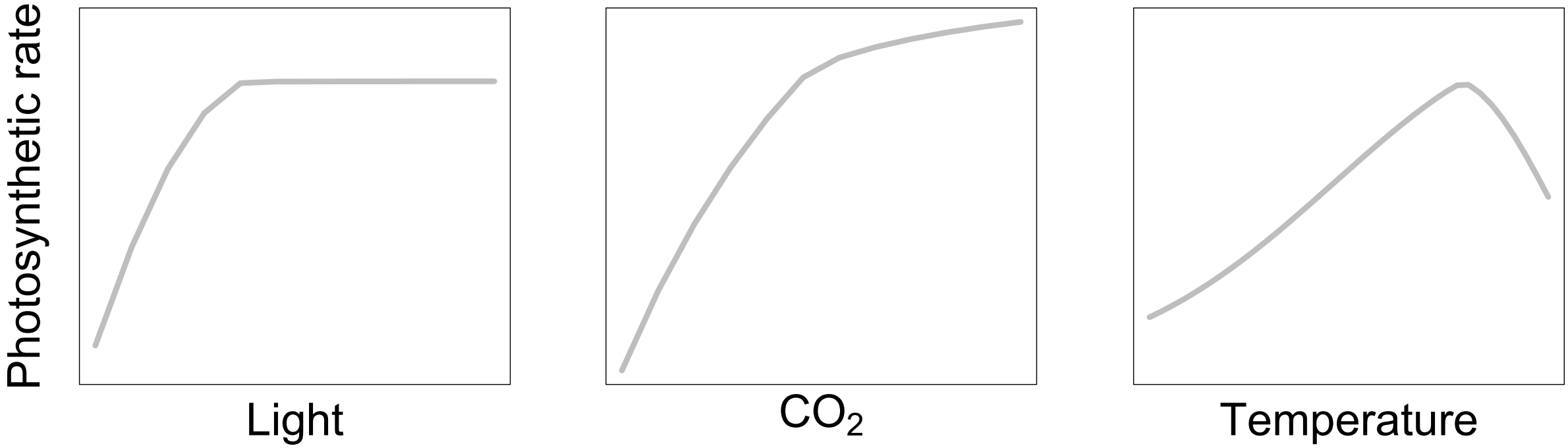
Planta
© by Springer-Verlag 1980

A Biochemical Model of Photosynthetic CO_2 Assimilation in Leaves of C_3 Species

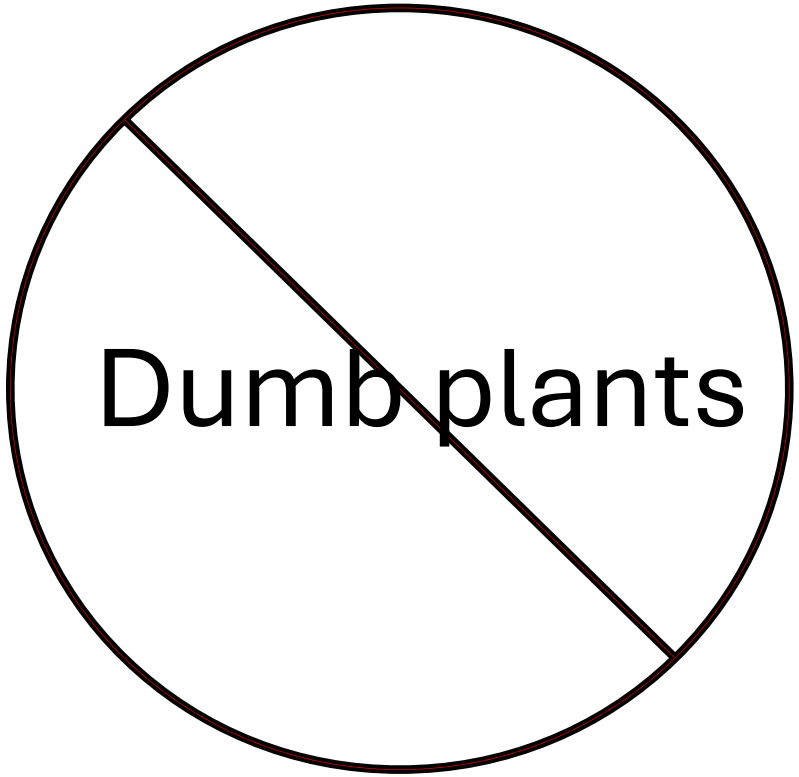
G.D. Farquhar[1], S. von Caemmerer[1], and J.A. Berry[2]

[1] Department of Environmental Biology, Research School of Biological Sciences, Australian National University, P.O. Box 475, Canberra City ACT 2601, Australia and
[2] Carnegie Institution of Washington, Department of Plant Biology, Stanford, Cal. 94305, USA

These produce short term responses that match data

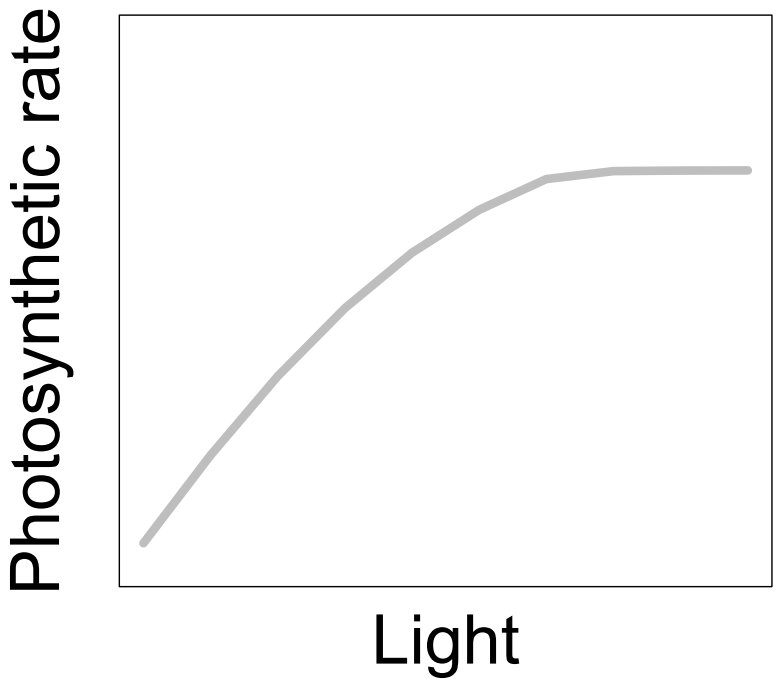


Why the uncertainty? Long-term responses differ from short-term responses due to acclimation

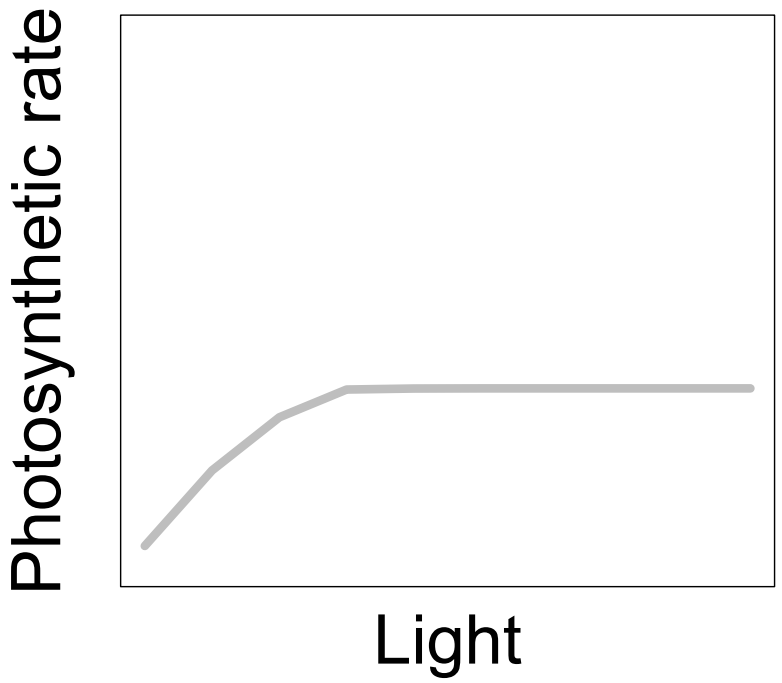

Dumb plants

Acclimation: an example

Acclimated to high light

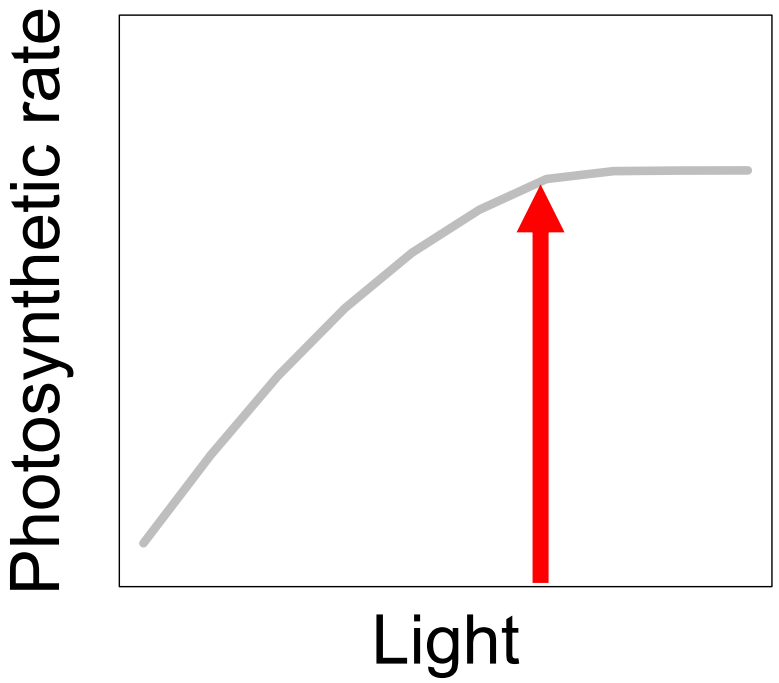


Acclimated to low light

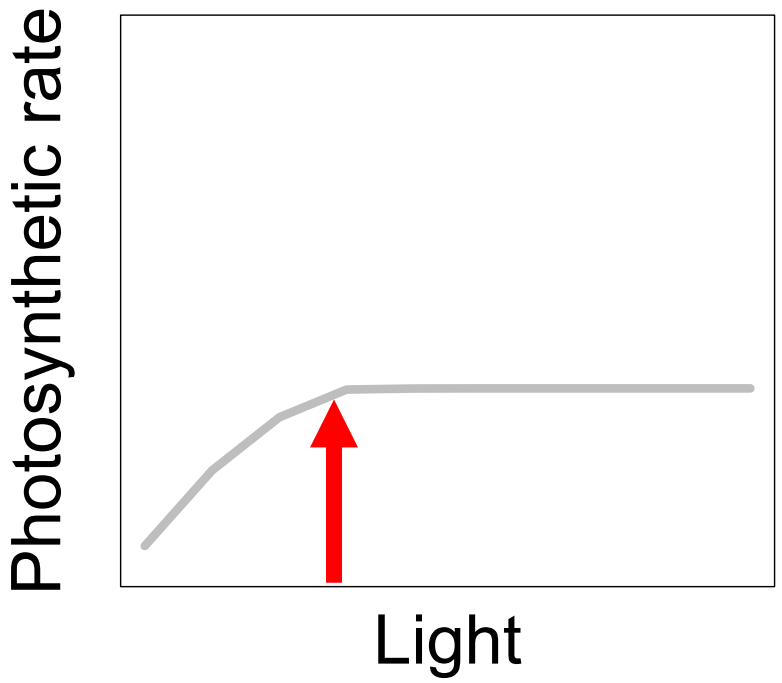


Acclimation: an example

Acclimated to high light



Acclimated to low light



Photosynthesis peaks a light level close to the level the leaf is acclimated to

Acclimation is ubiquitous and well known…

CO_2: Bazzaz (1990)

Annu. Rev. Ecol. Syst. 1990. 21:167–96
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THE RESPONSE OF NATURAL ECOSYSTEMS TO THE RISING GLOBAL CO_2 LEVELS

F. A. Bazzaz

Light: Boardman (1977)

Ann. Rev. Plant Physiol. 1977. 28:355–77
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COMPARATIVE PHOTOSYNTHESIS OF SUN AND SHADE PLANTS

N. K. Boardman
Division of Plant Industry, CSIRO, Canberra City, A.C.T. 2601, Australia

Temperature: Berry & Björkman (1980)

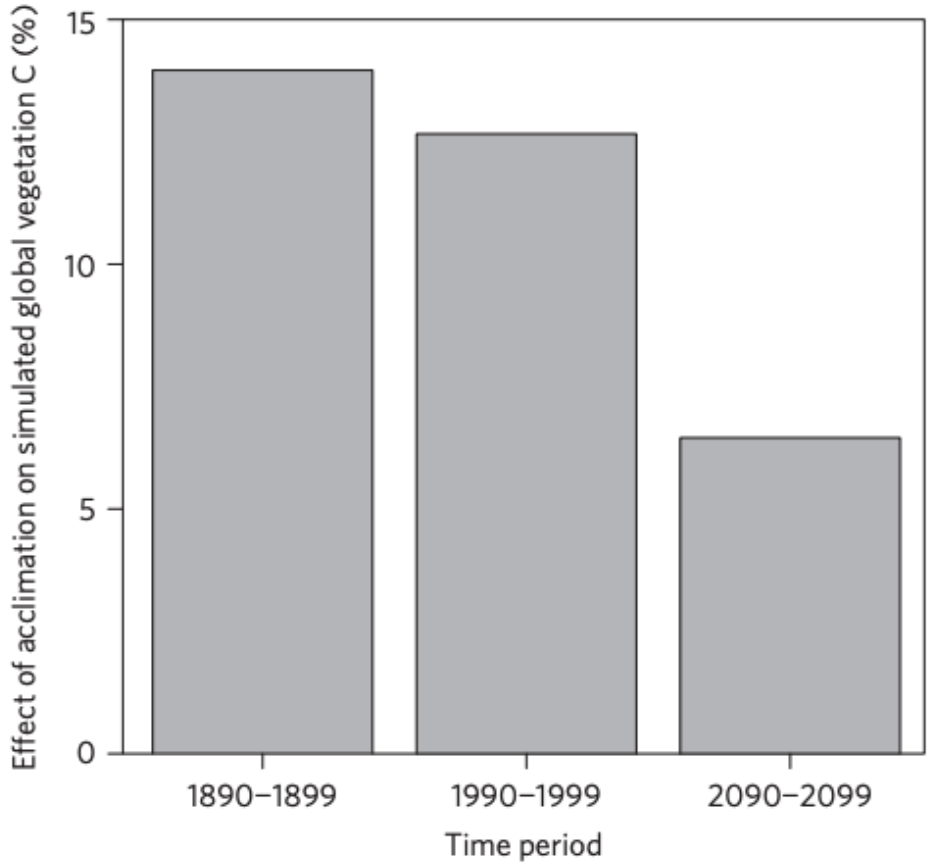
Ann. Rev. Plant Physiol. 1980. 31:491–543
Copyright © 1980 by Annual Reviews Inc. All rights reserved

PHOTOSYNTHETIC RESPONSE AND ADAPTATION TO TEMPERATURE IN HIGHER PLANTS

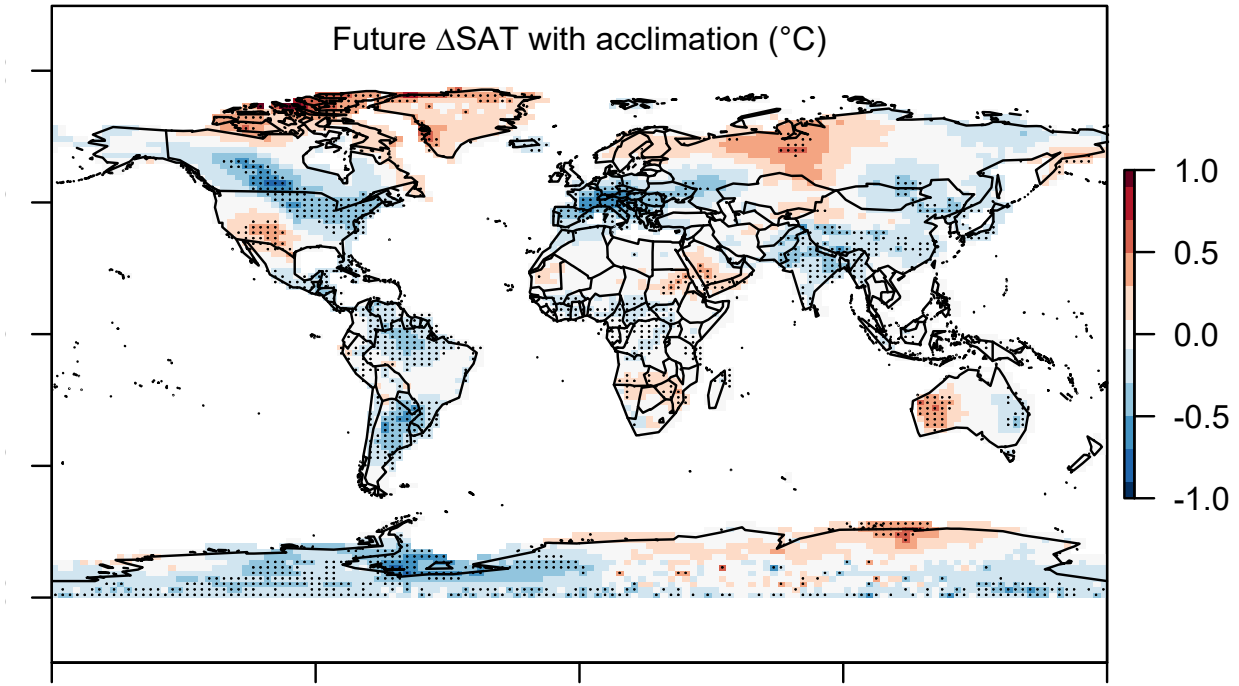
Joseph Berry and Olle Björkman[1]

…and can impact carbon cycling and climate

Acclimation increase future C storage by ~6%



Acclimation alters future temperature by >1°C



Smith et al. (2017) *Nature Climate Change*

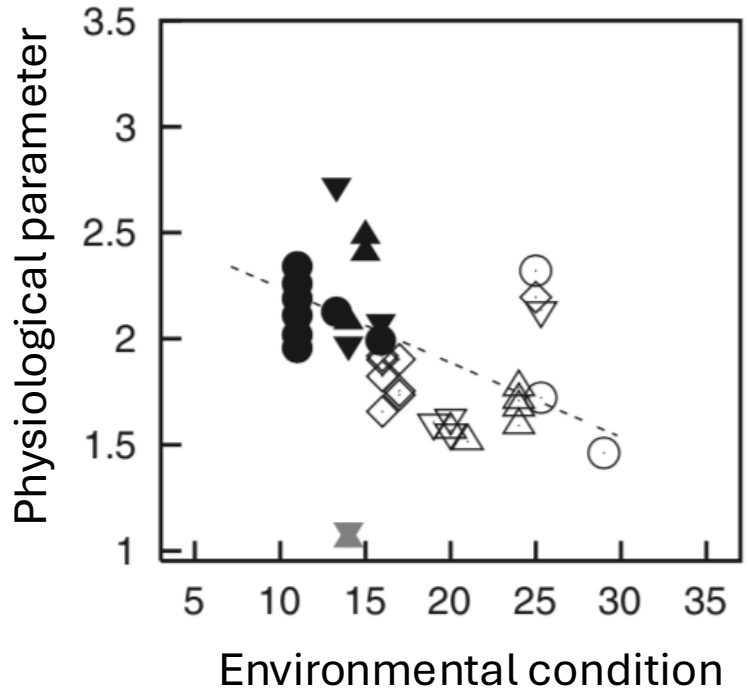
Smith et al. (2017) *JAMES*

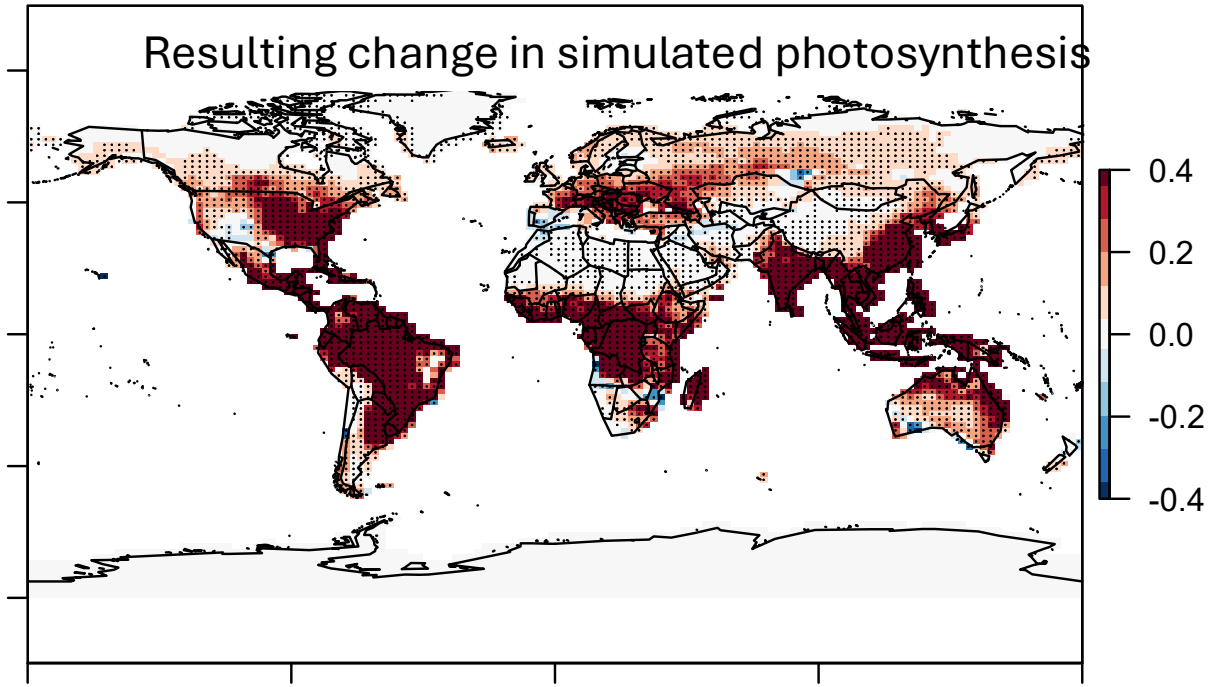
Unfortunately,
no theoretical model for
photosynthetic acclimation exists

Lack of theory results in...

Lack of theory results in…

- Unreliable future predictions (overparameterization, tuning)
  - Reliance on statistical models



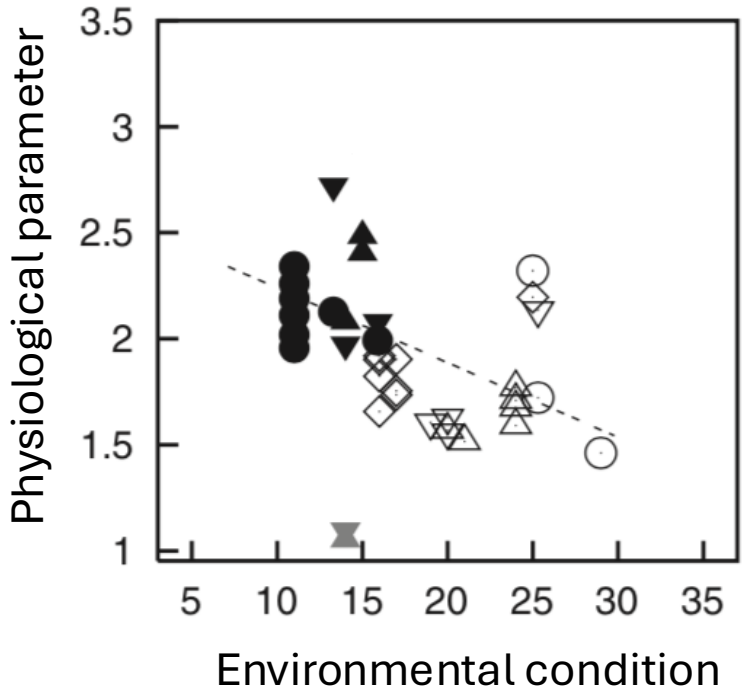


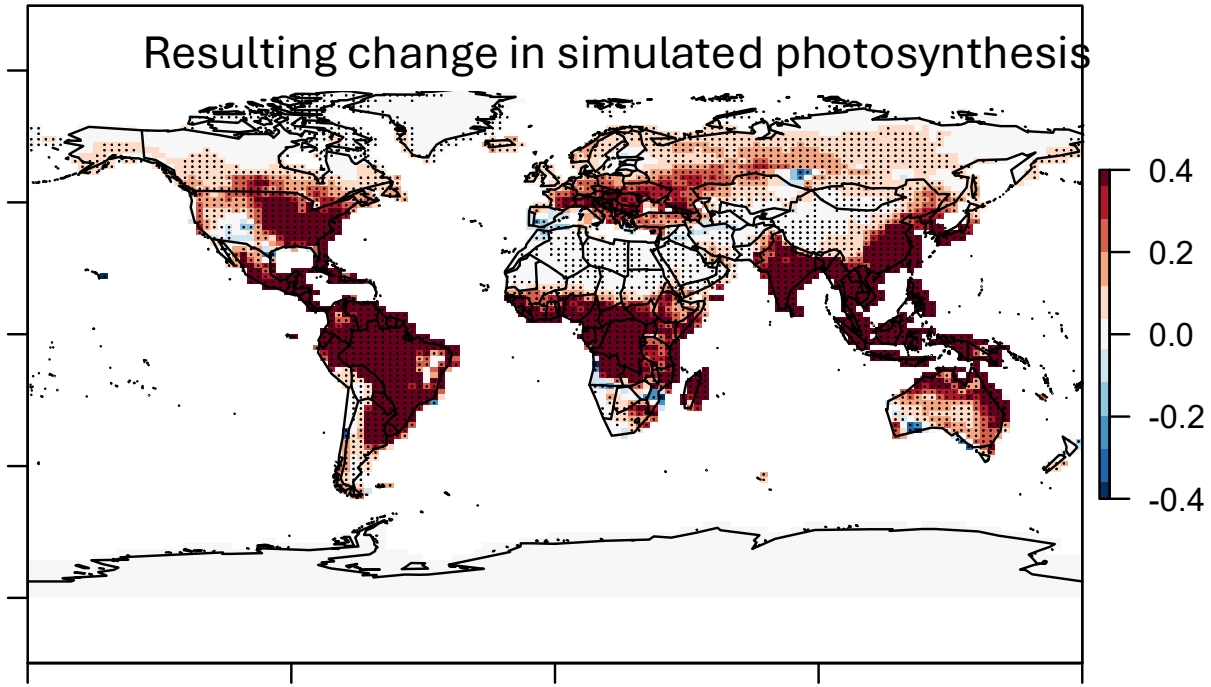
Kattge and Knorr (2007) *PC&E*

Smith et al. (2017) *JAMES*

Lack of theory results in…

- Inability to test mechanisms





Kattge and Knorr (2007) *PC&E*

Smith et al. (2017) *JAMES*

La

- In

Global Change Biology

REPORT | Open Access

Increasing the spatial and temporal impact of ecological research: A roadmap for integrating a novel terrestrial process into an Earth system model

Emily Kyker-Snowman, Danica L. Lombardozzi, Gordon B. Bonan, Susan J. Cheng, Jeffrey S. Dukes, Serita D. Frey, Elin M. Jacobs, Risa McNellis, Joshua M. Rady, Nicholas G. Smith ... See all authors

First published: 20 September 2021 | https://doi.org/10.1111/gcb.15894 | Citations: 27

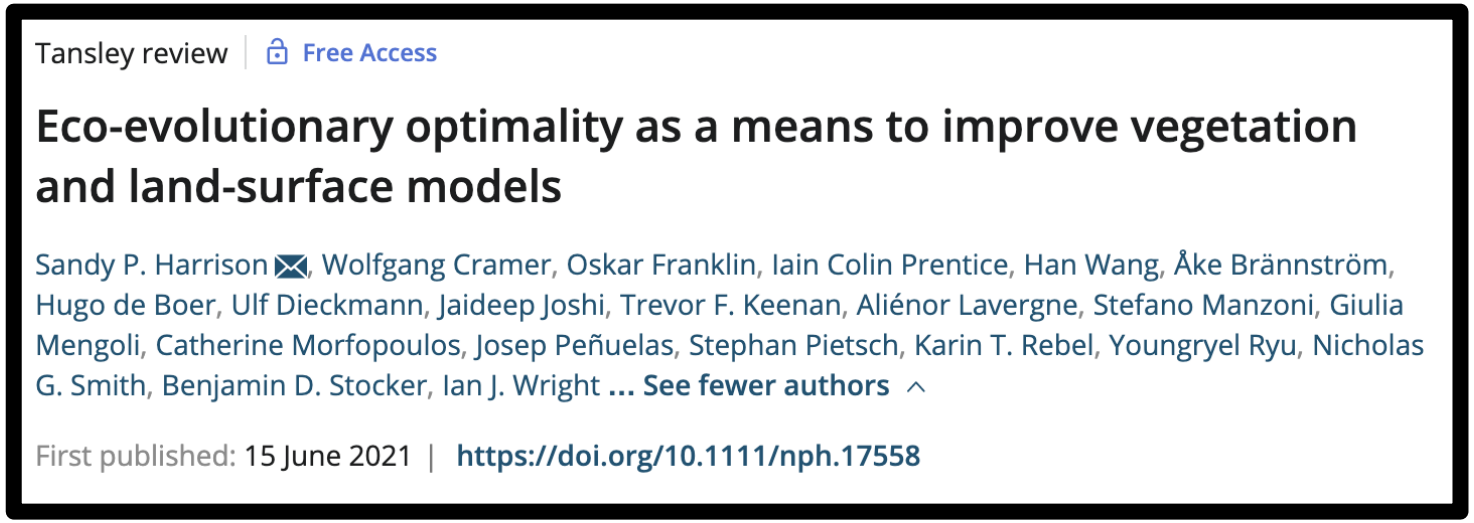
See also commentary on this article by Moore, 28, 343–345



Kyker-Snowman et al. (2022) *Glob Change Biol*

So, we developed a mechanistic model of photosynthetic acclimation

Based on **optimization** and the **first principles** of plant physiological theory



Tansley review | Free Access

Eco-evolutionary optimality as a means to improve vegetation and land-surface models

Sandy P. Harrison ✉, Wolfgang Cramer, Oskar Franklin, Iain Colin Prentice, Han Wang, Åke Brännström, Hugo de Boer, Ulf Dieckmann, Jaideep Joshi, Trevor F. Keenan, Aliénor Lavergne, Stefano Manzoni, Giulia Mengoli, Catherine Morfopoulos, Josep Peñuelas, Stephan Pietsch, Karin T. Rebel, Youngryel Ryu, Nicholas G. Smith, Benjamin D. Stocker, Ian J. Wright **... See fewer authors** ^

First published: 15 June 2021 | https://doi.org/10.1111/nph.17558

Harrison et al. (2021) *New Phyt*

Optimization: Least cost theory

Optimally, plants will maintain fastest rate of photosynthesis at the lowest summed resource cost (water and nutrient use)

Wright et al. (2003) *Am Naturalist*

Optimal photosynthesis

Photosynthesis = *f*{stomatal conductance, photosynthetic biochemistry}

Optimal photosynthesis

Photosynthesis = *f*{stomatal conductance,
photosynthetic biochemistry}



Must predict optimal rates of both

Optimal photosynthesis

Photosynthesis = *f*{**stomatal conductance**, photosynthetic biochemistry}

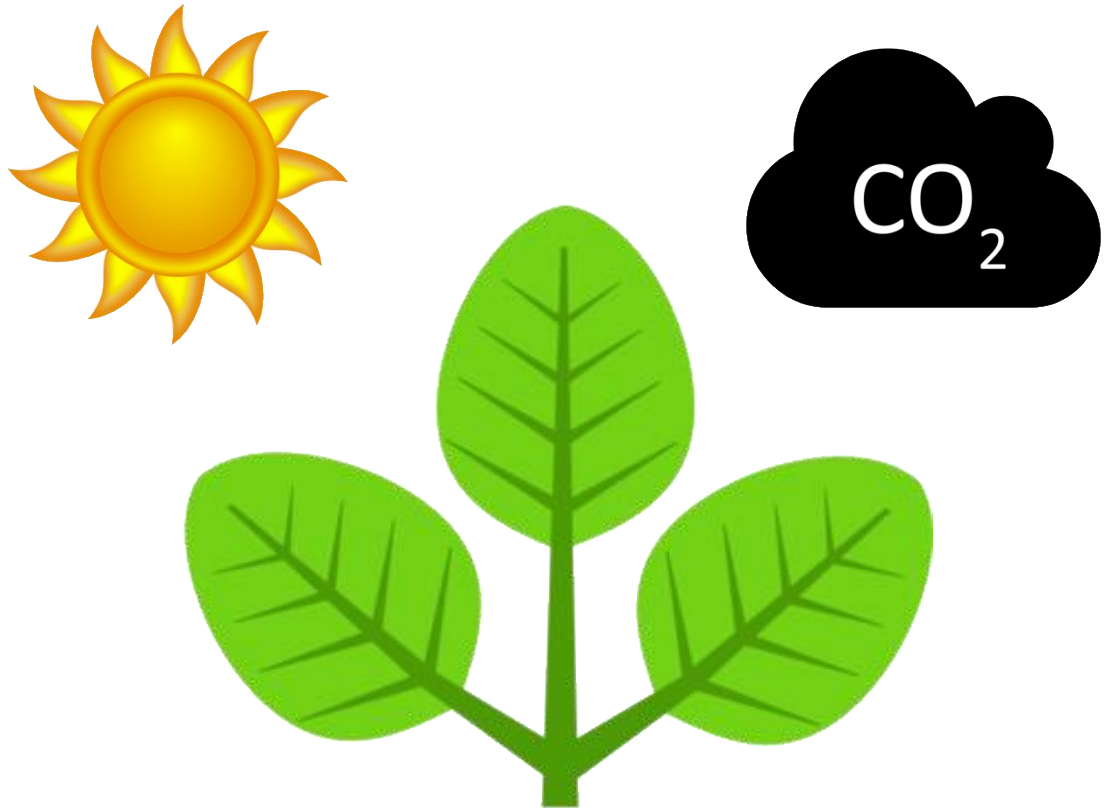
[TALK TO ME LATER ABOUT THIS IF YOU ARE INTERESTED]

Prentice et al. (2014) *Ecol Letters*; Wang et al. (2017) *Nature Plants*; Smith et al. (2019) *Ecol Letters*; Smith and Keenan (2020) *Glob Change Biol*

Optimal photosynthesis

Photosynthesis = *f*{stomatal conductance,
photosynthetic biochemistry}

Biochemistry optimization

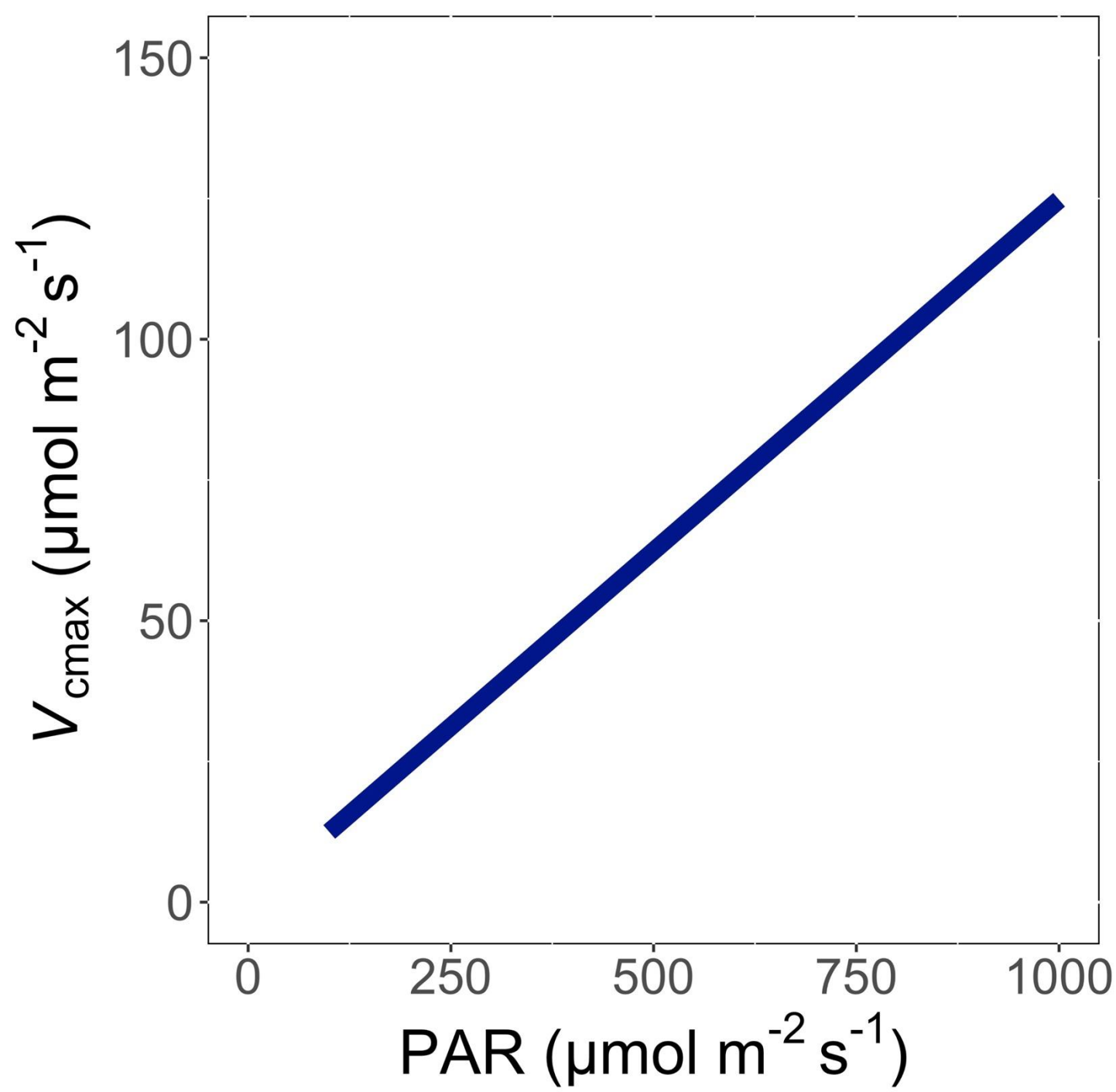


Plant biochemistry setup will aim for equal limitation by all factors

[TALK TO ME LATER ABOUT THE MATHS]

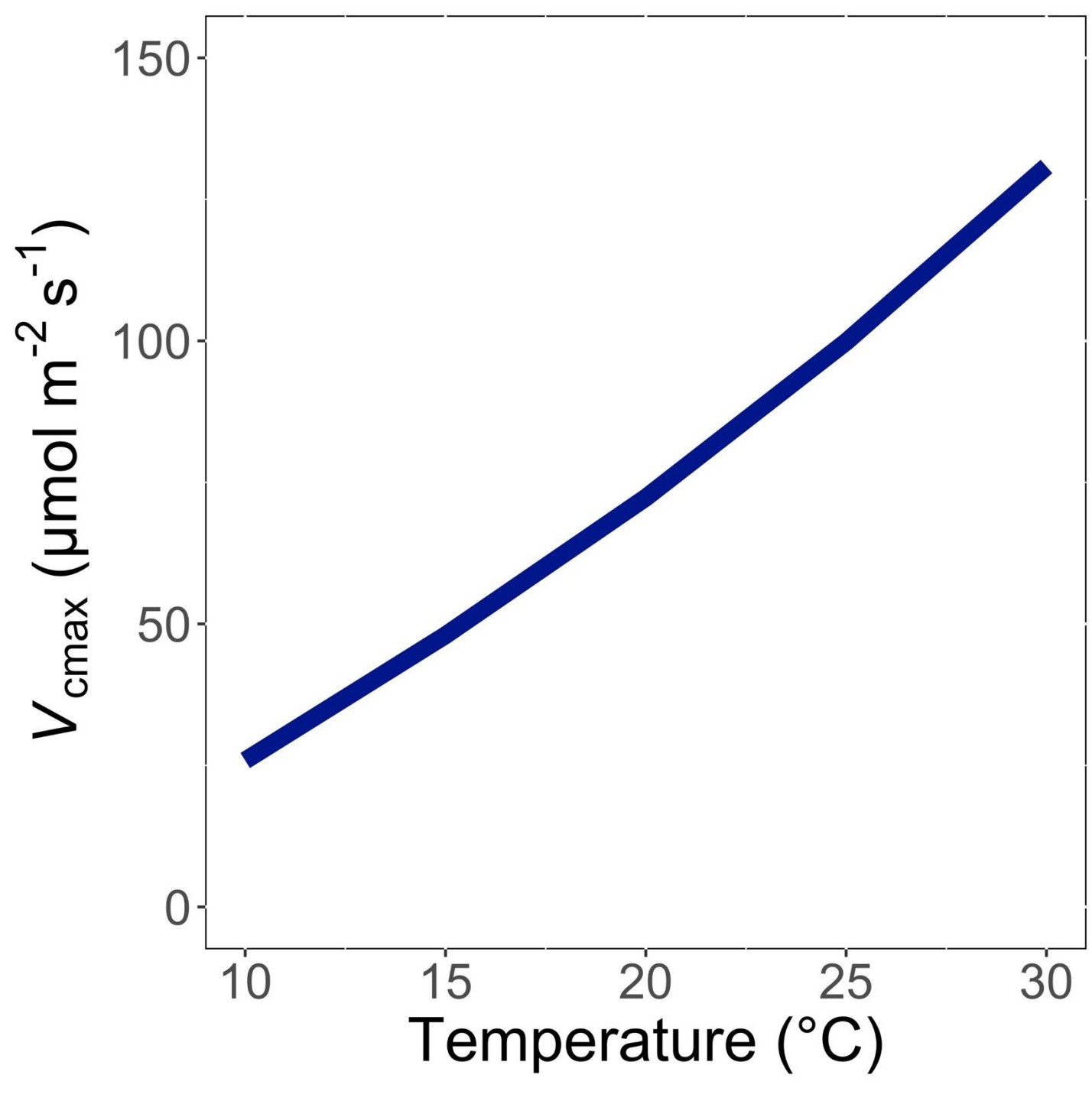
Biochemistry trait

The maximum rate of Rubisco carboxylation (V_{cmax})



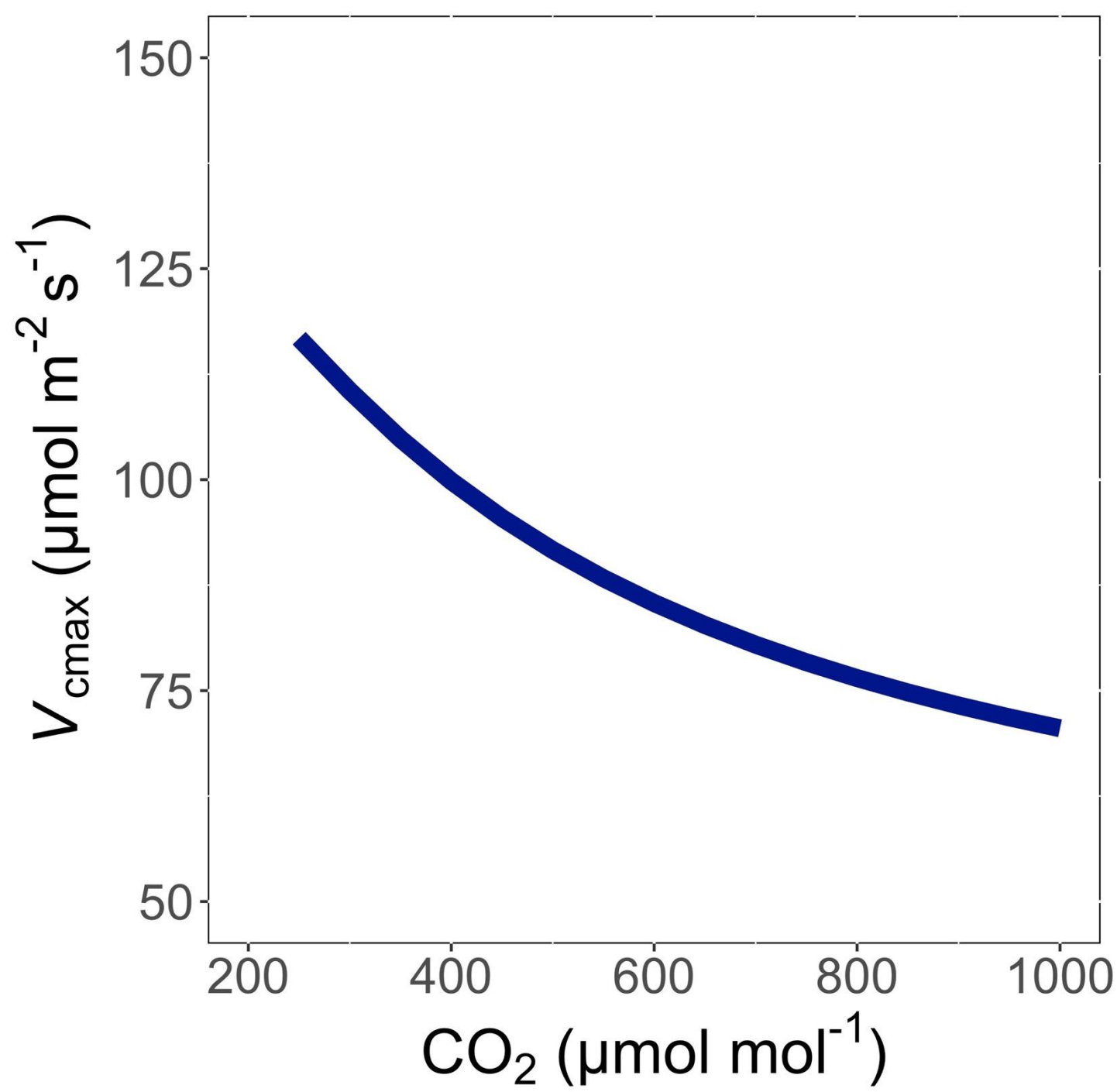
V_{cmax} increases with light because of greater electron transport

Smith et al. (2019) *Ecol Letters*; Smith and Keenan (2020) *Glob Change Biology*



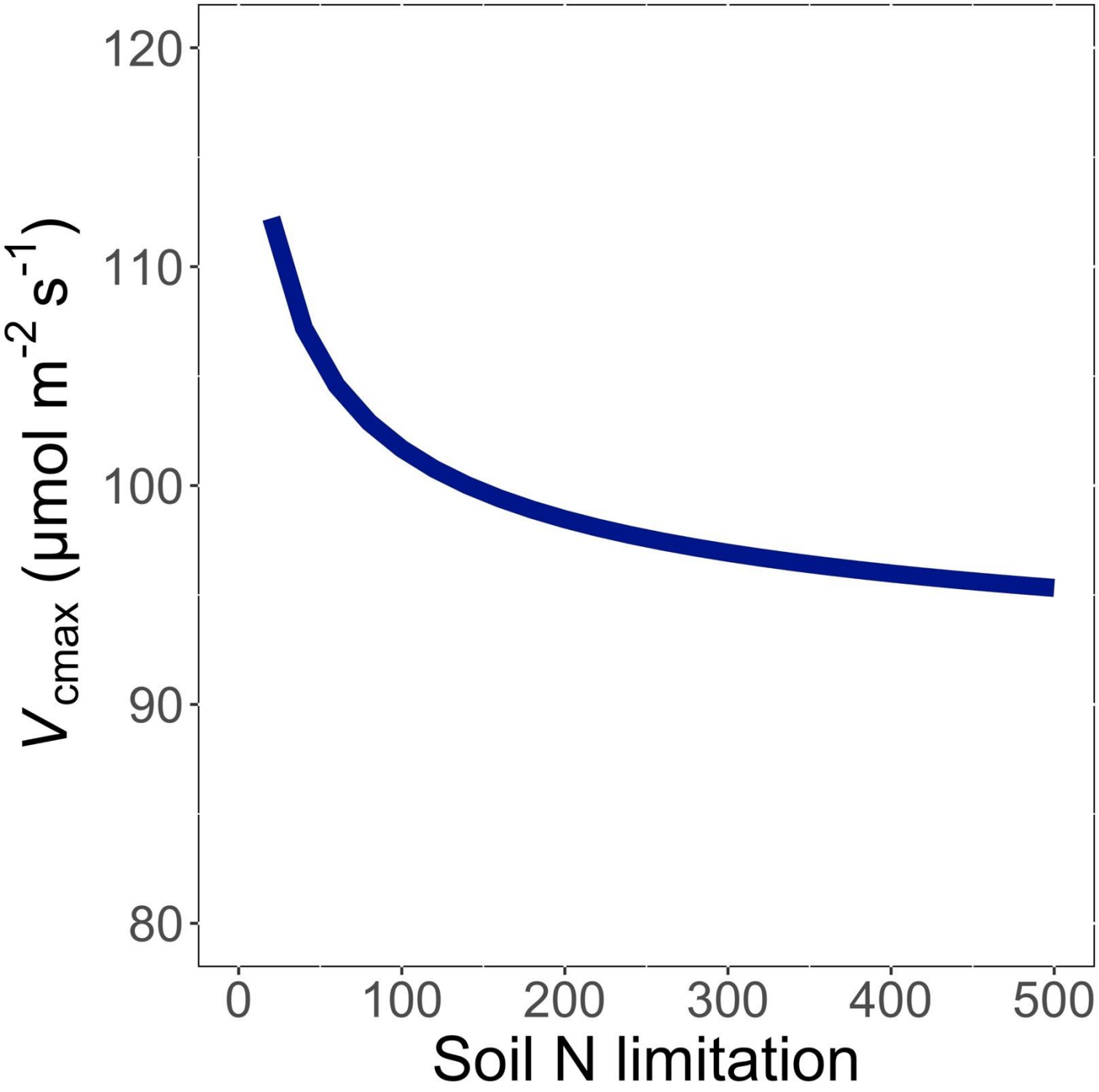
V_{cmax} increases with temperature because of greater electron transport and photorespiration

Smith et al. (2019) *Ecol Letters*; Smith and Keenan (2020) *Glob Change Biology*



V_{cmax} decreases with CO_2 because of greater CO_2 in the leaf and less photorespiration

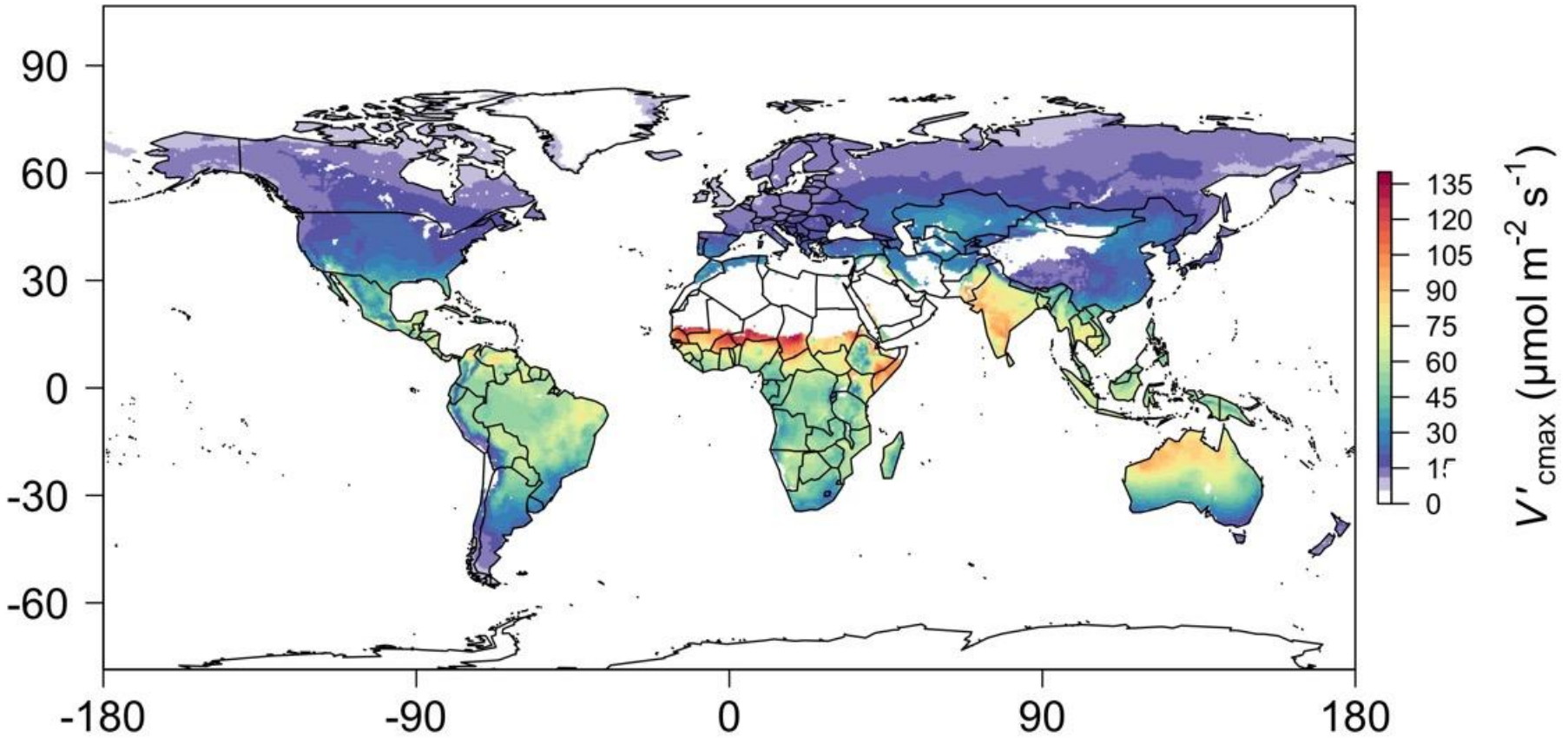
Smith et al. (2019) *Ecol Letters*; Smith and Keenan (2020) *Glob Change Biology*



V_{cmax} decreases with soil N limitation because Rubisco requires a lot of N

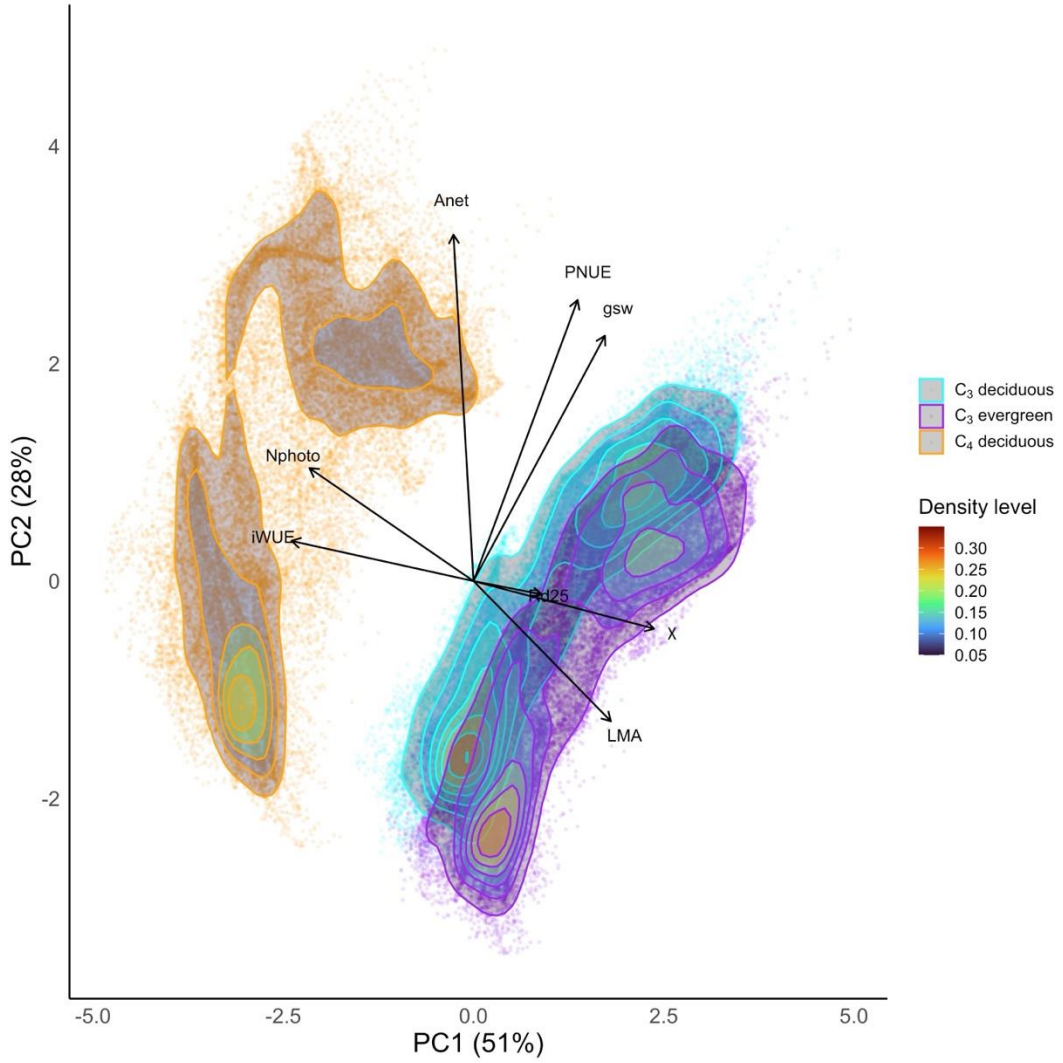
Smith et al. (2019) *Ecol Letters*; Smith and Keenan (2020) *Glob Change Biology*

We can predict optimal traits in different environments



Smith et al. (2019) *Ecology Letters*

We can even predict optimal trait strategies



Smith (in review)

Ok, great, but now what?

We can use the theory as a null model to explore acclimation mechanisms

Let’s tackle some big questions in plant ecophysiology!

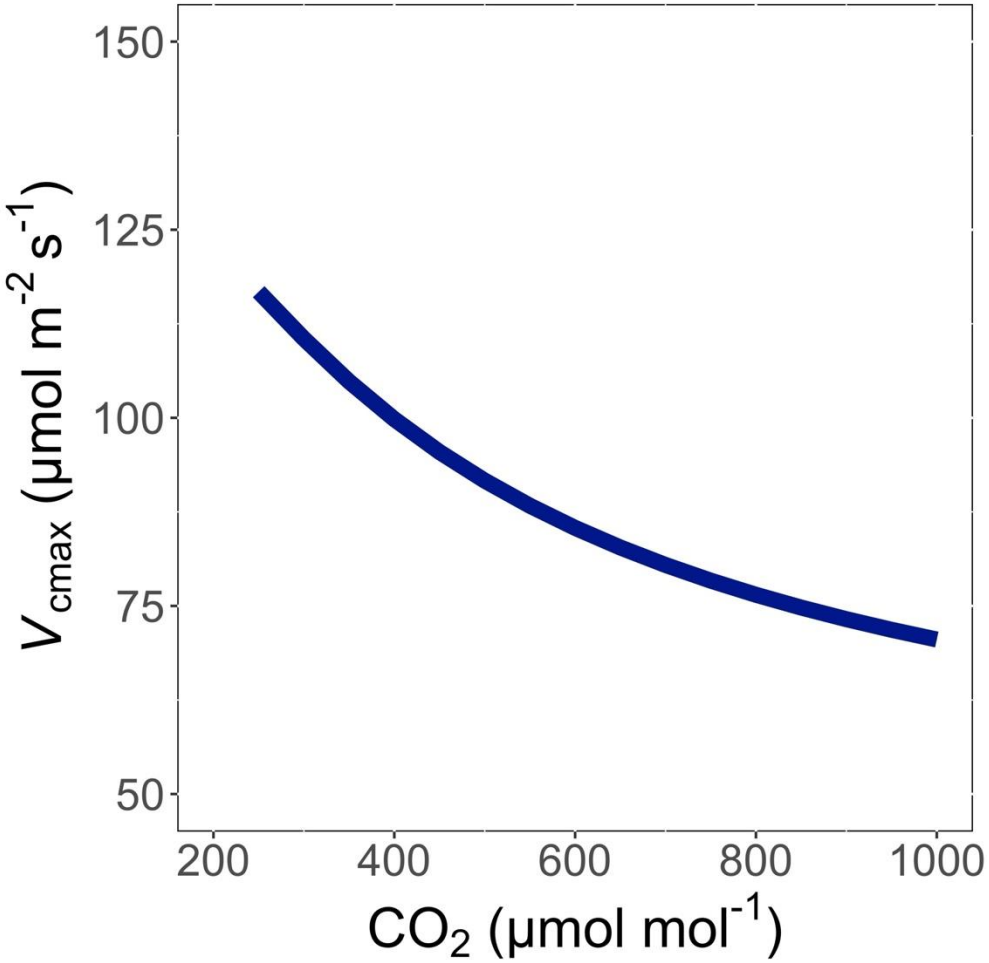
Big questions

1. Is photosynthesis optimized to the environment?
2. How do plants acclimate to soil nitrogen?
3. What does acclimation mean for future terrestrial biogeochemical cycling?
4. When is C_4 photosynthesis an advantage over C_3 photosynthesis?

Big questions

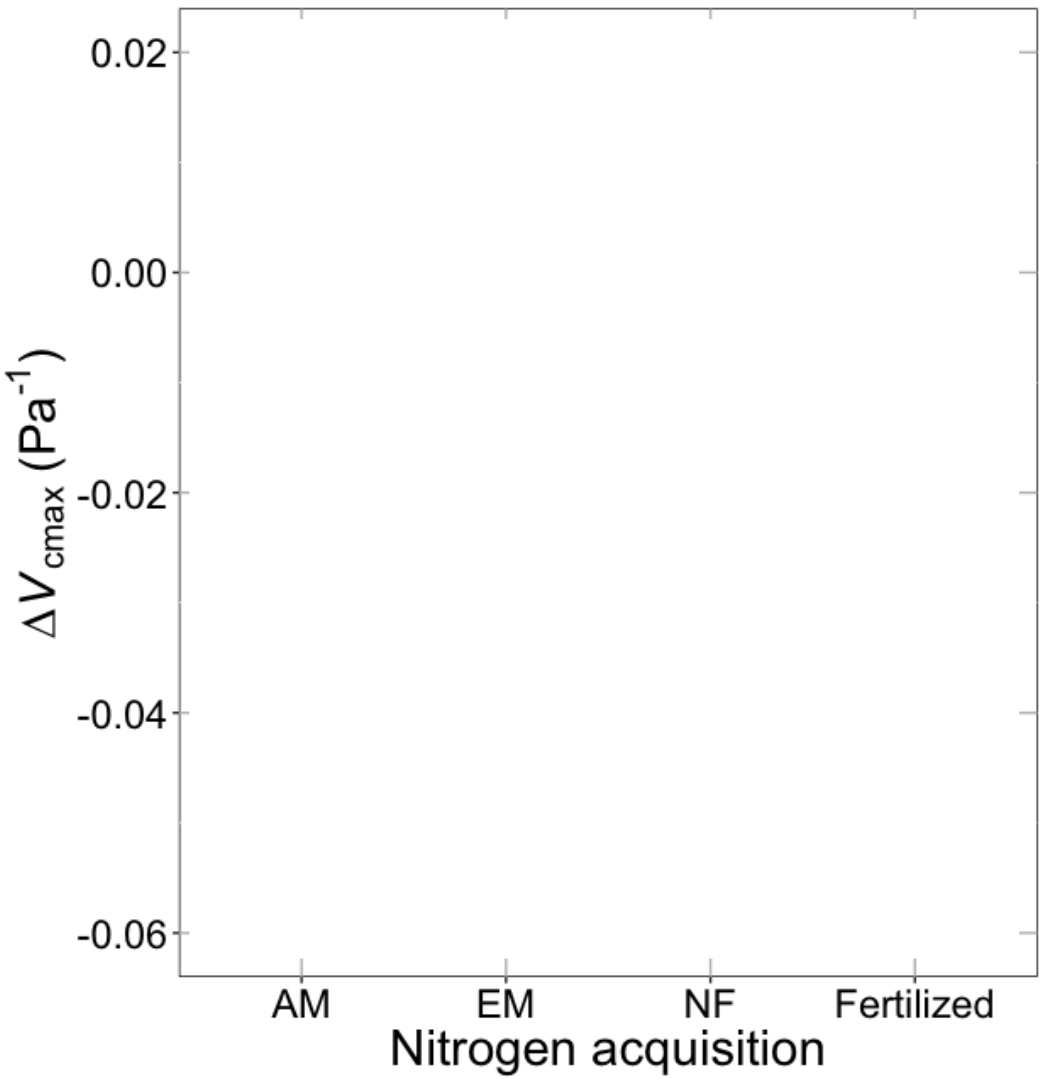
1. ~~Is photosynthesis optimized to the environment?~~
2. ~~How do plants acclimate to soil nitrogen?~~
3. What does acclimation mean for future terrestrial biogeochemical cycling? A CO_2 story
4. ~~When is C_4 photosynthesis an advantage over C_3 photosynthesis?~~

Expected future responses

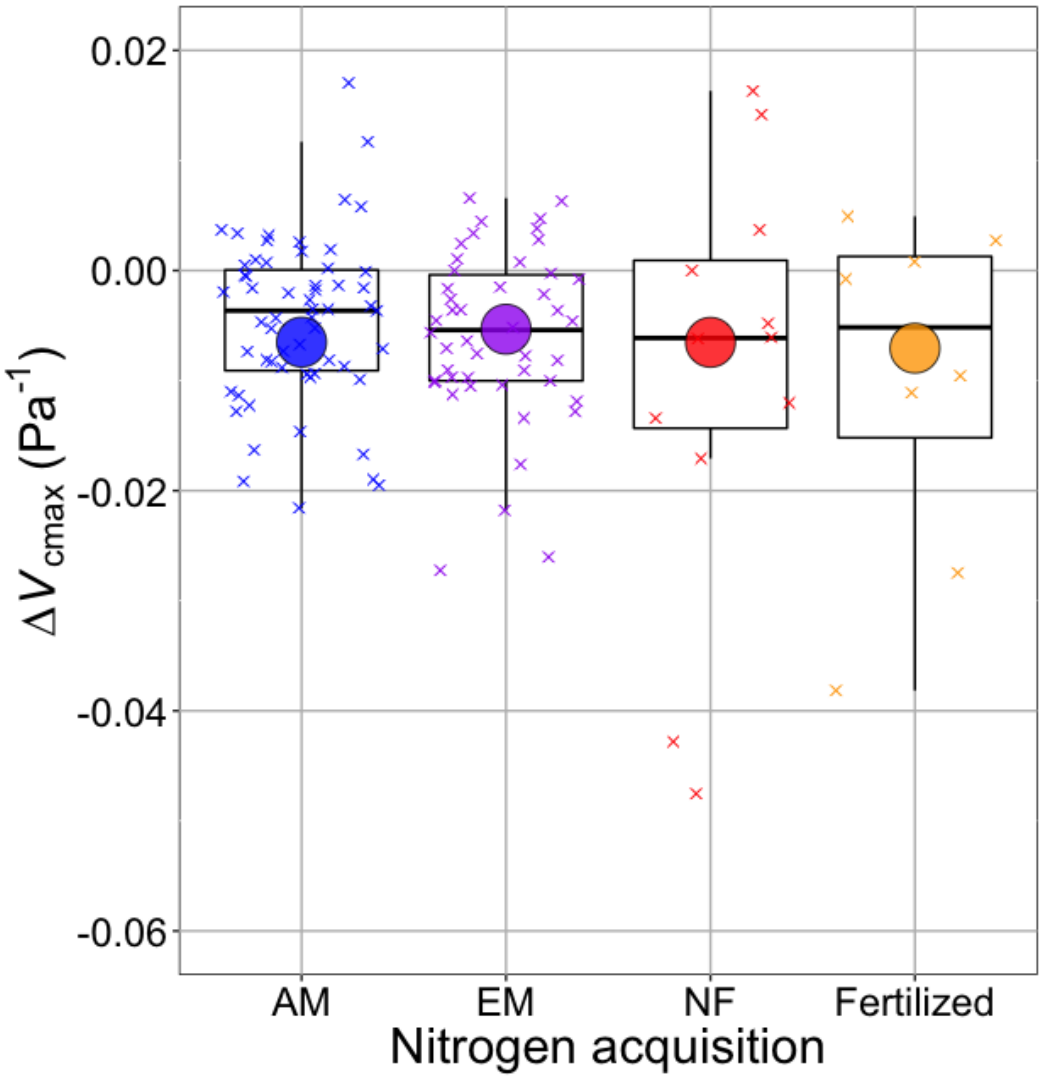


V_{cmax} decreases with CO_2 because of greater CO_2 in the leaf and less photorespiration

Smith et al. (2019) *Ecol Letters*



Smith and Keenan (2020) *Glob Change Biol*

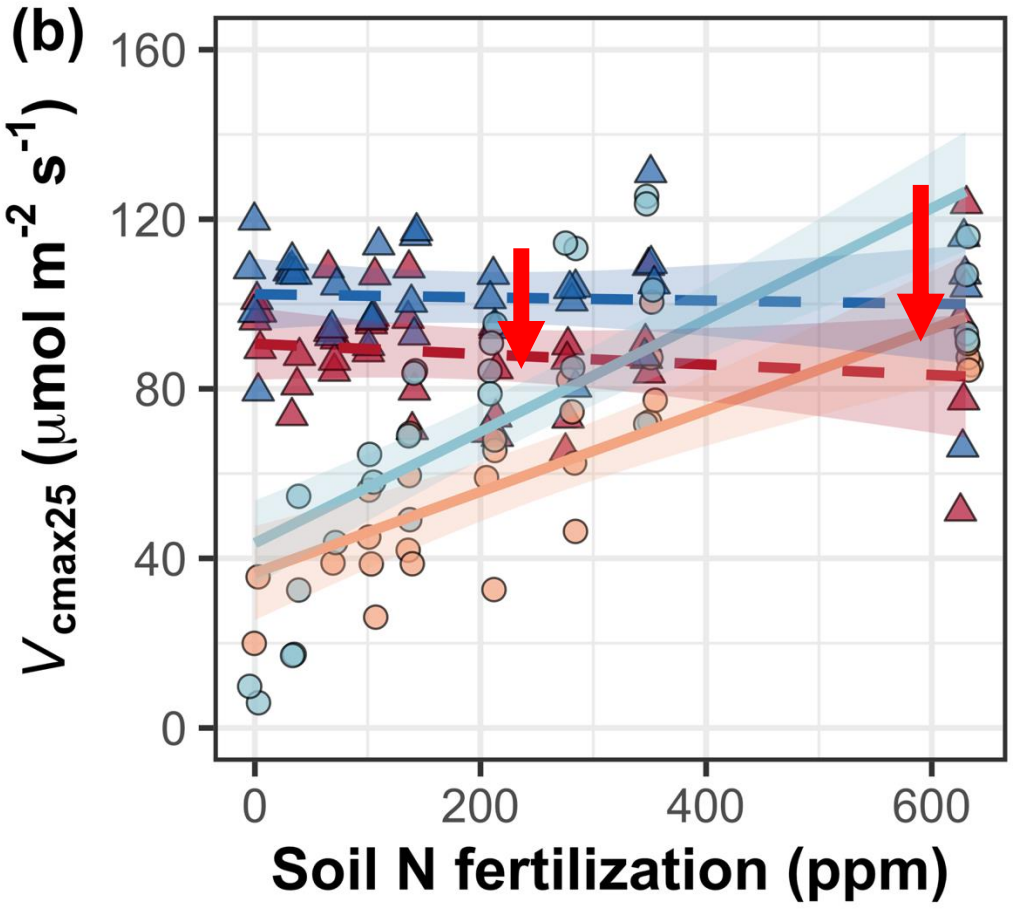


V_{cmax} changes with CO_2 in ways expected from optimization

Boxes = data = -0.0063 Pa^{-1}

Circles = predicted = -0.0066 Pa^{-1}

Smith and Keenan (2020) *Glob Change Biol*



We further confirmed this universality of this response under 18 nitrogen by inoculation treatments

Treatment

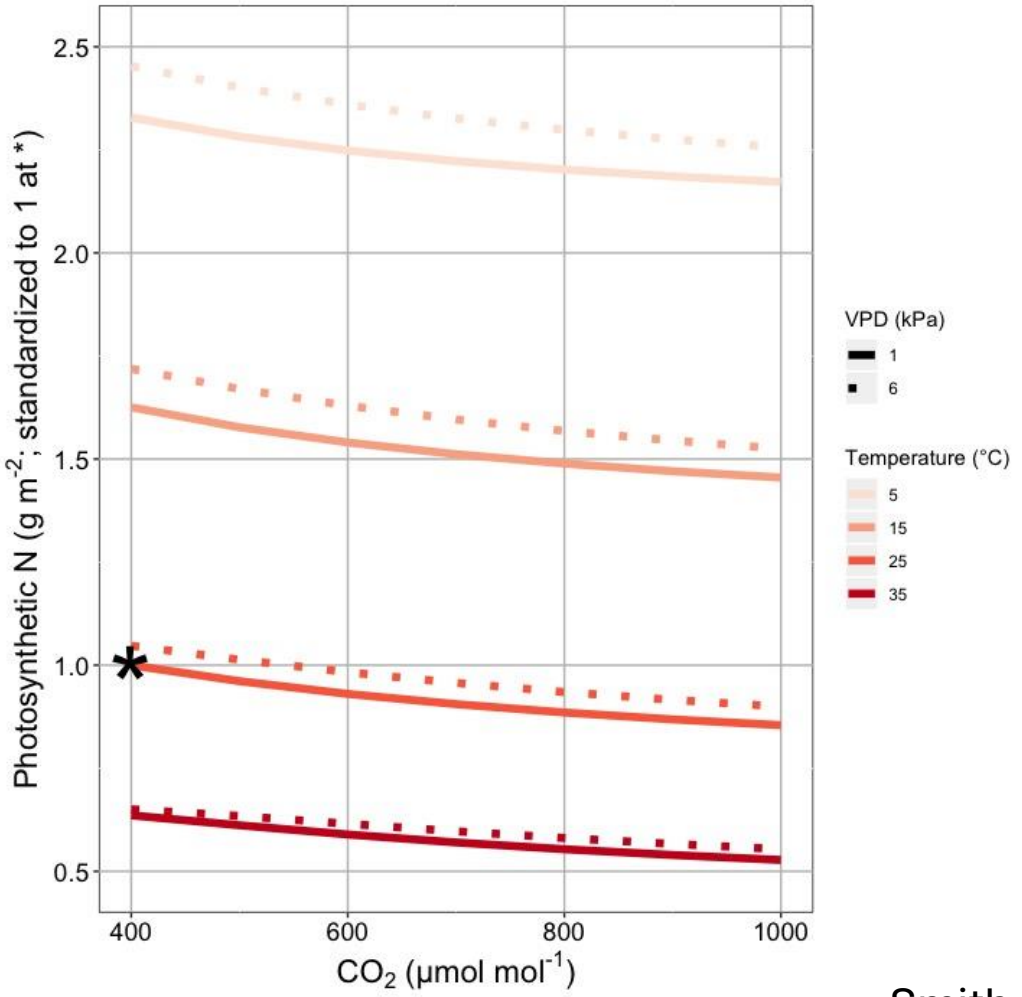
- Elevated CO_2, inoculated
- Elevated CO_2, uninoculated
- Ambient CO_2, inoculated
- Ambient CO_2, uninoculated





Perkowski, Ezekannagha, and Smith (2025) *J Exp Botany*

This generally suggests lower nitrogen demand under future conditions

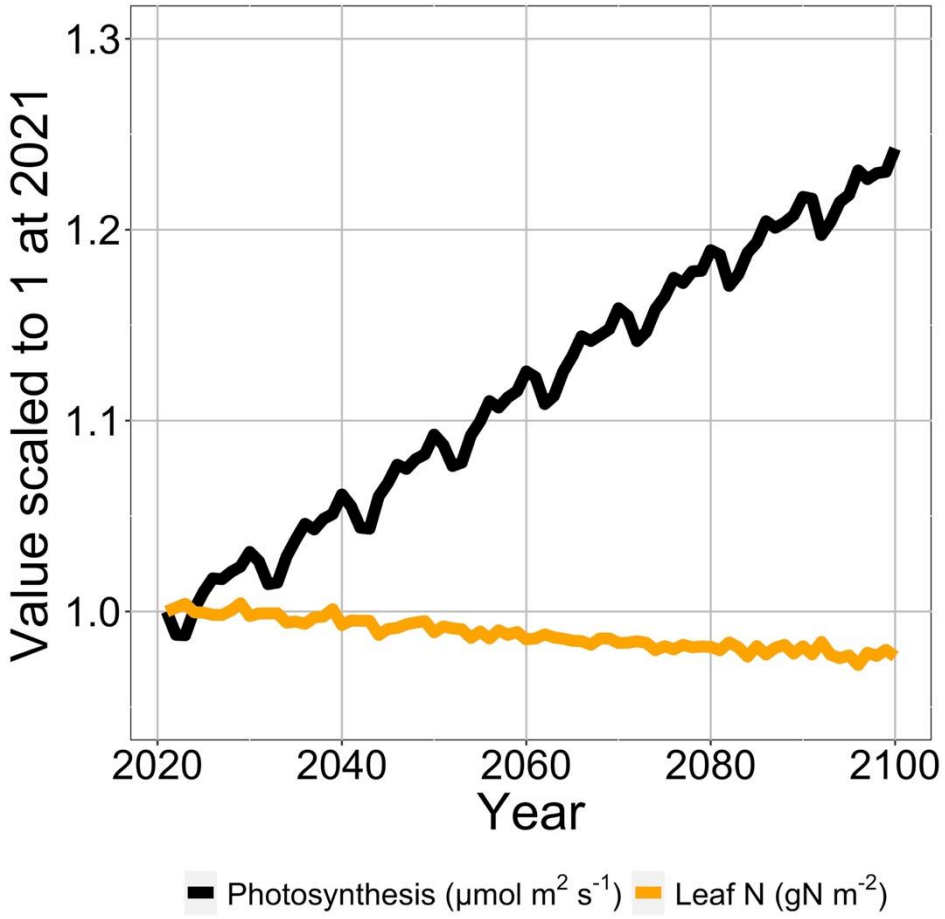


Smith and Keenan (2020) *Glob Change Biol*

Let’s run a model out into the future!

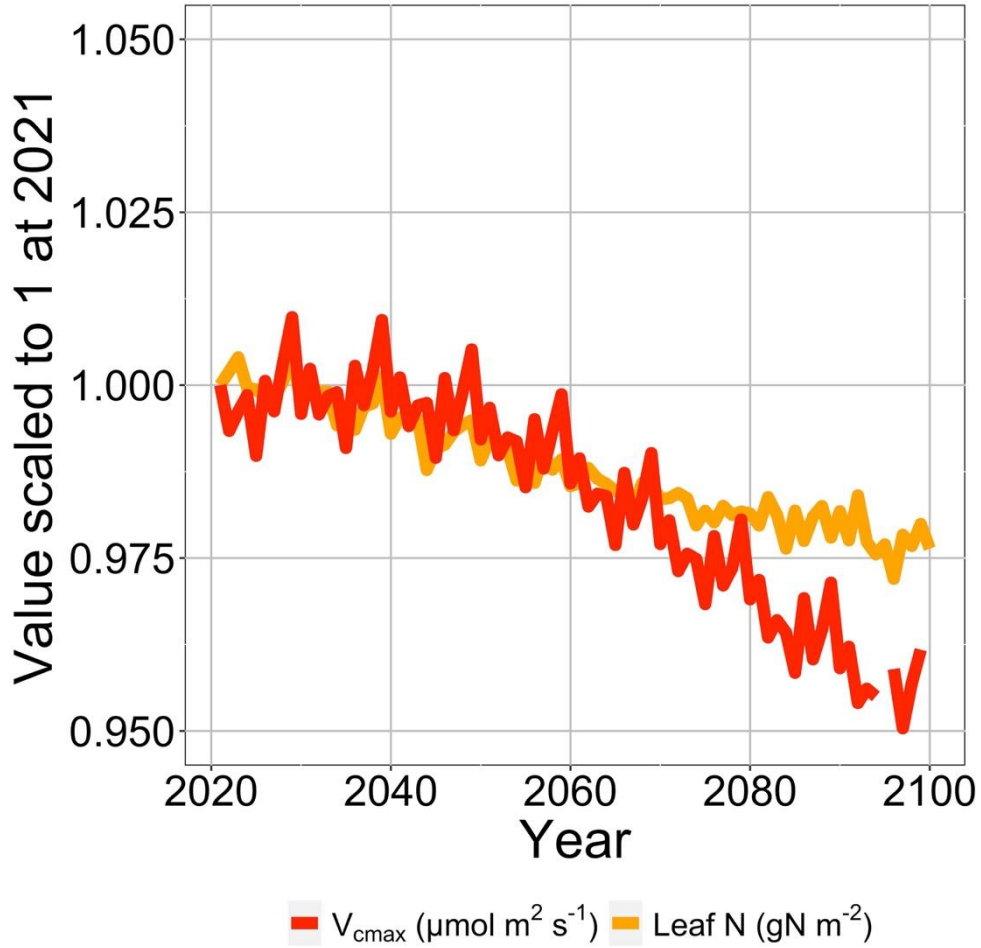


Photosynthesis increases with elevated CO_2 at lower leaf N



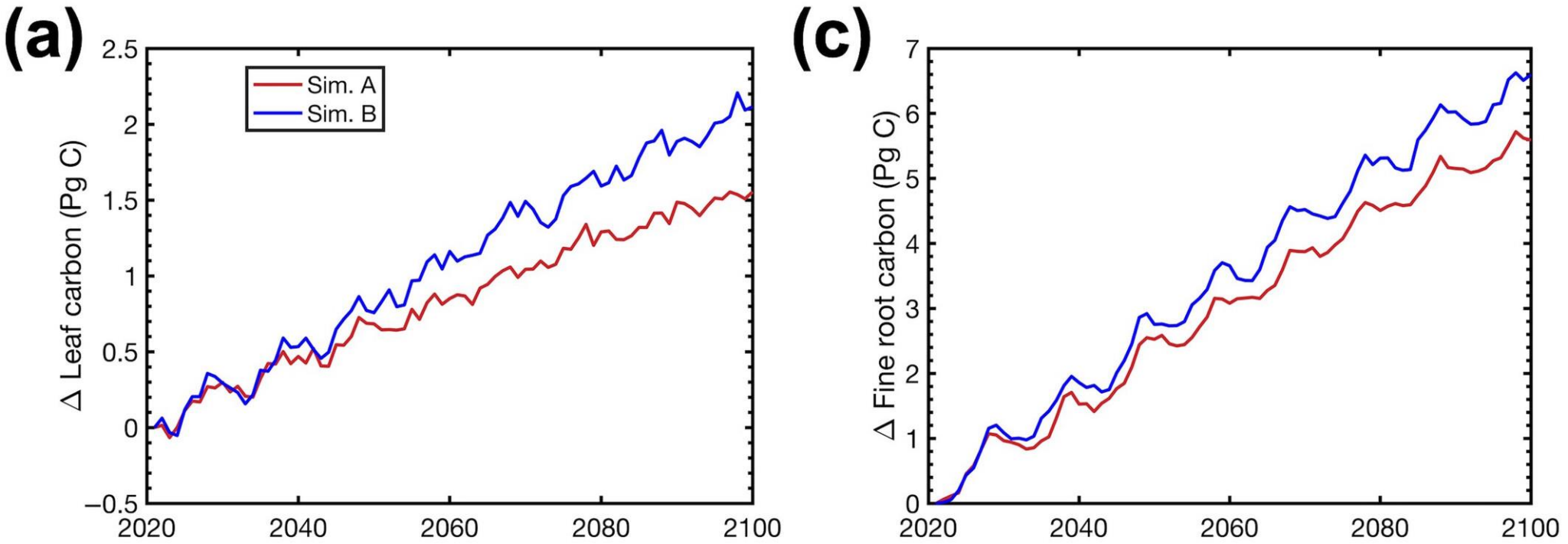
Smith et al. (2024) *Glob Change Biol*

Leaf N reduction is due to a reduction in photosynthetic capacity



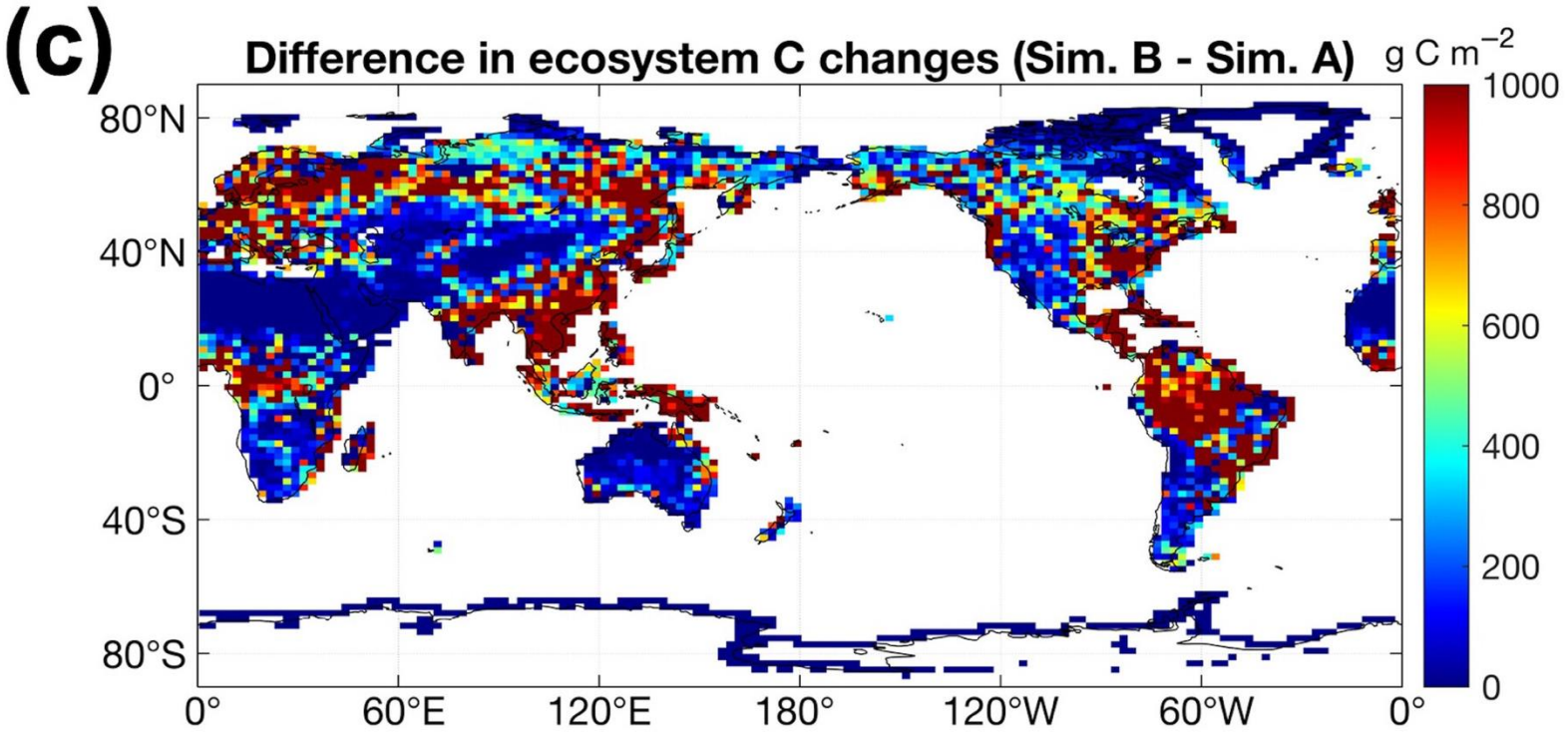
Smith et al. (2024) *Glob Change Biol*

Plants build more leaves and fine roots



Smith et al. (2024) *Glob Change Biol*

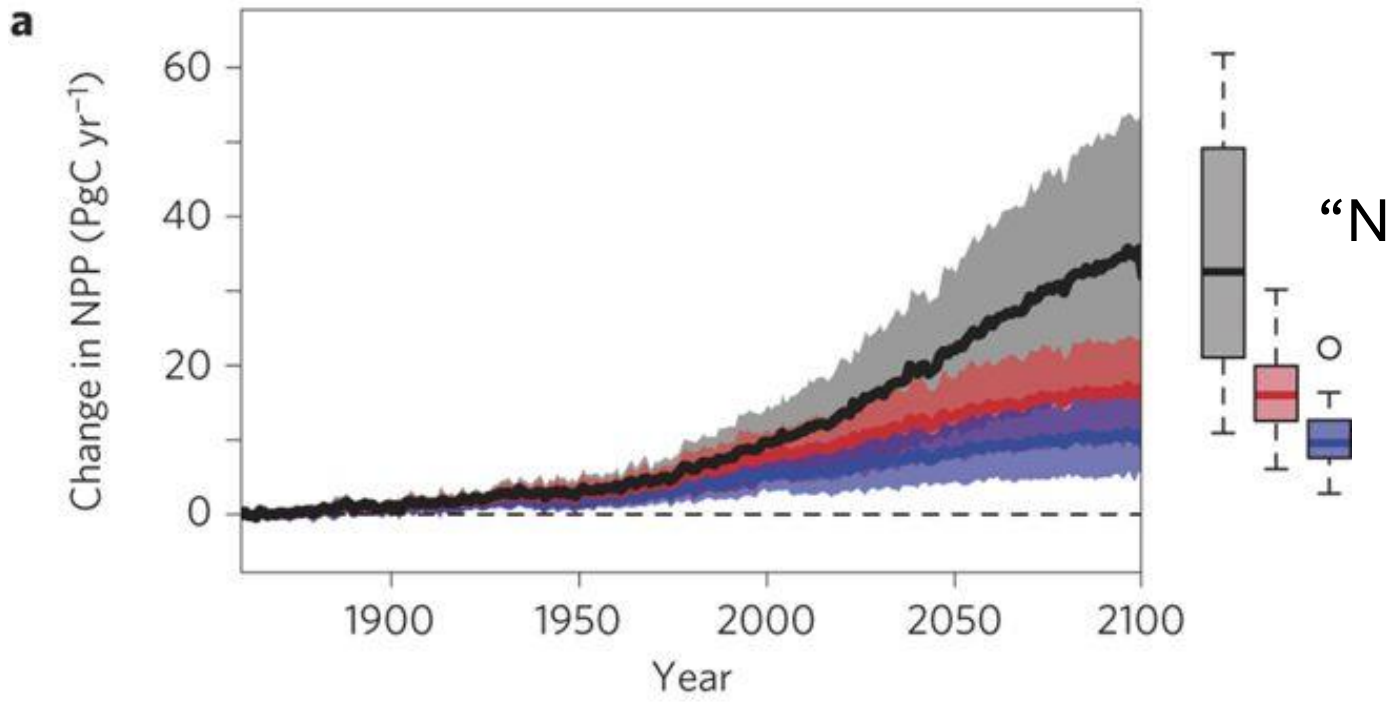
Leaf N savings increases long-term ecosystem carbon stocks



50% increase in ecosystem carbon from CO_2 acclimation

Smith et al. (2024) *Glob Change Biol*

Need to rethink nutrient limitation in models?



"No limitation" simulation

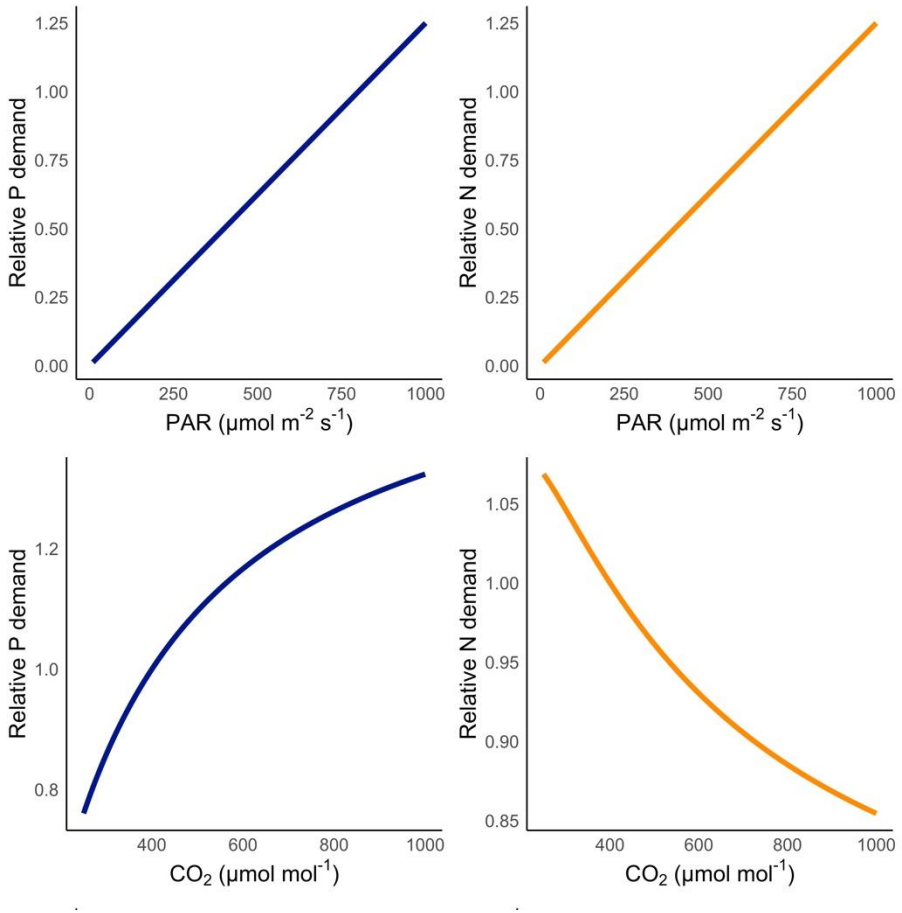
"Nutrient limitation" simulations

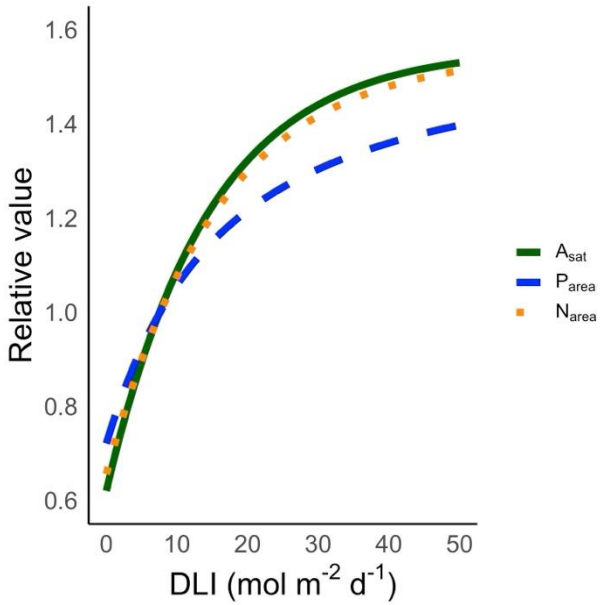
Weider et al. (2015)

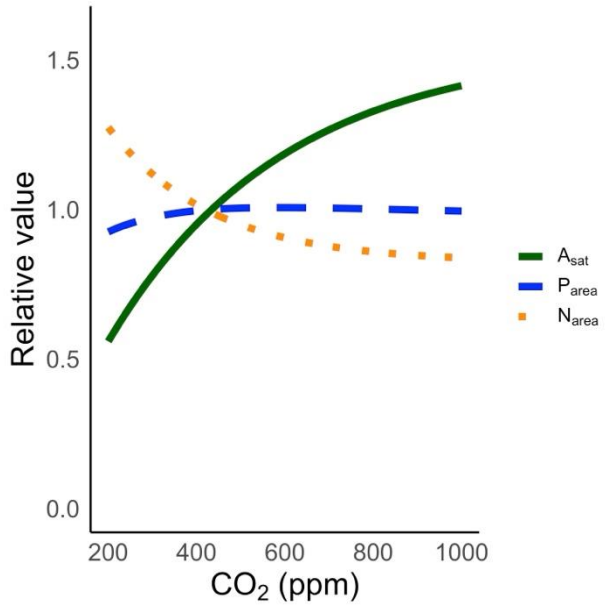
What does acclimation mean for future terrestrial biogeochemical cycling?

Photosynthetic acclimation to elevated CO_2 frees up nutrients that enhance ecosystem carbon uptake and storage beyond what is currently simulated in models

Looking forward: a simple theory for understanding photosynthesis-phosphorus relationships

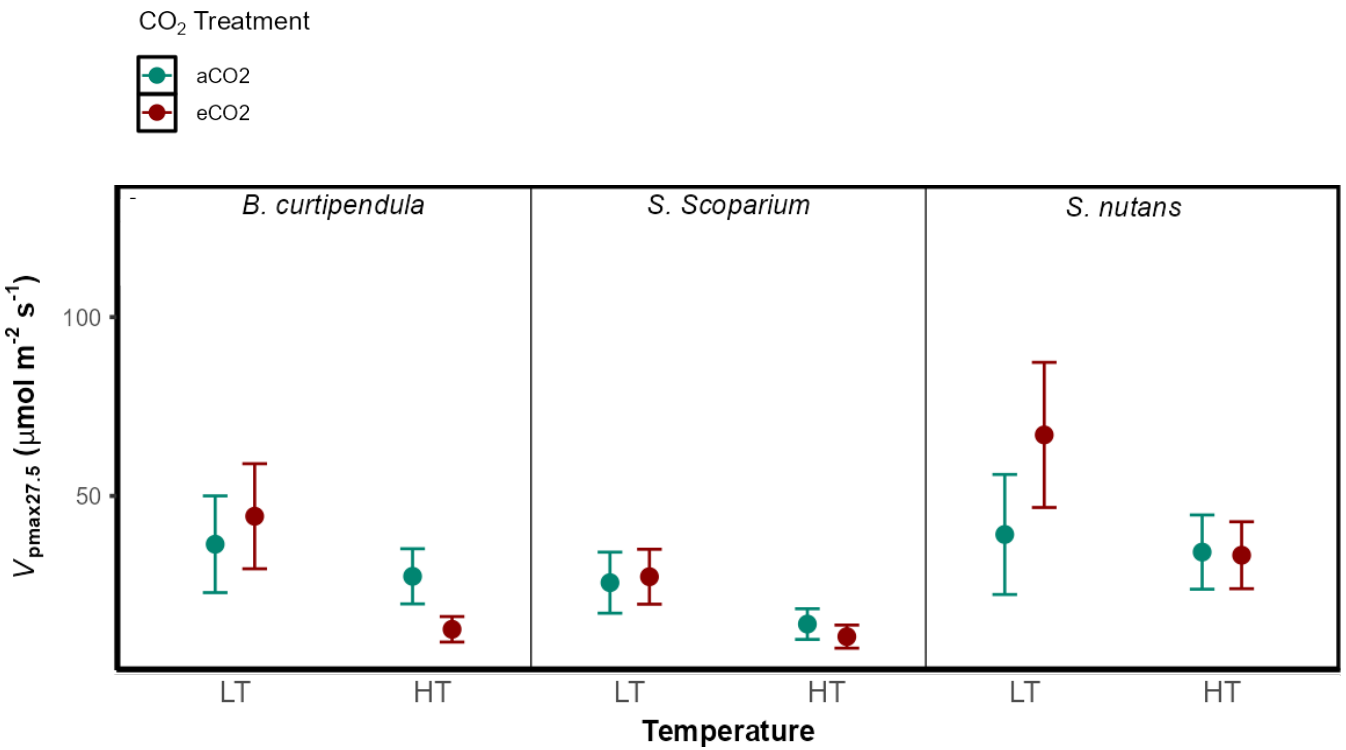




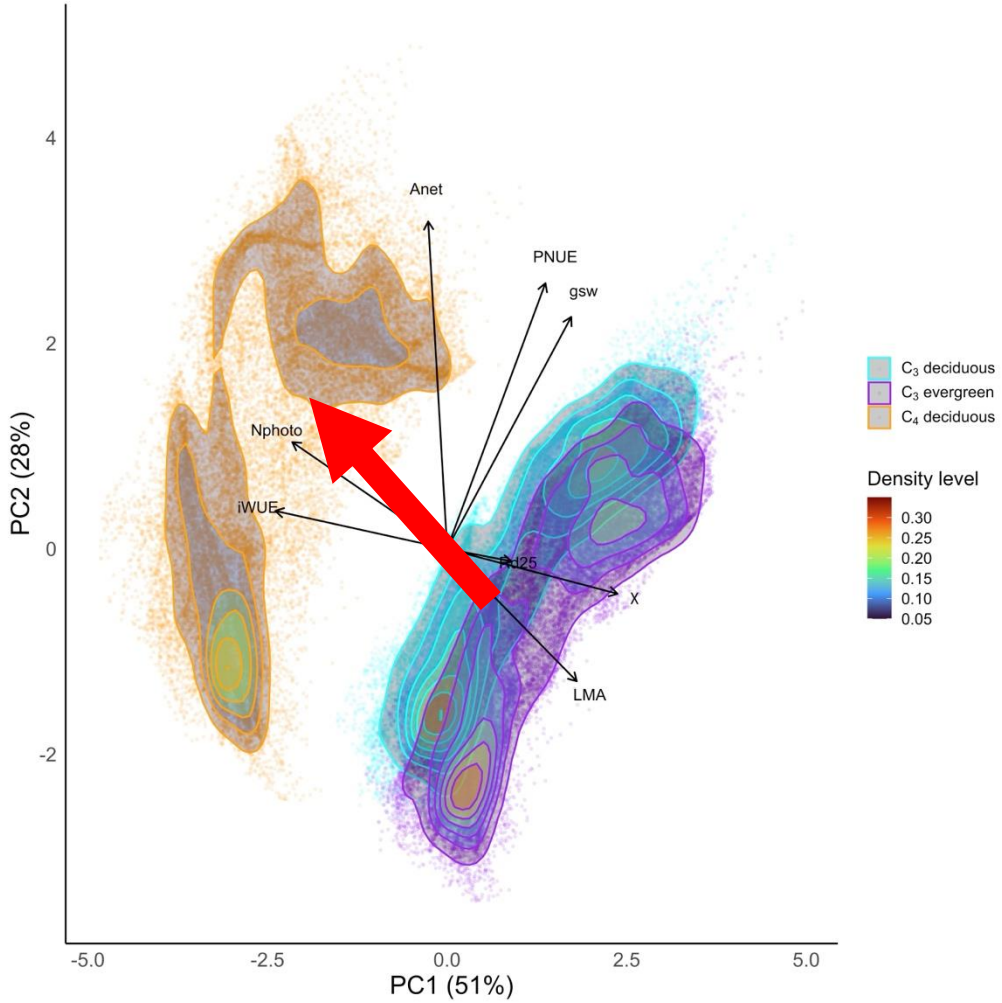


Smith et al. (in prep)

Looking forward: integrating C_4 optimization



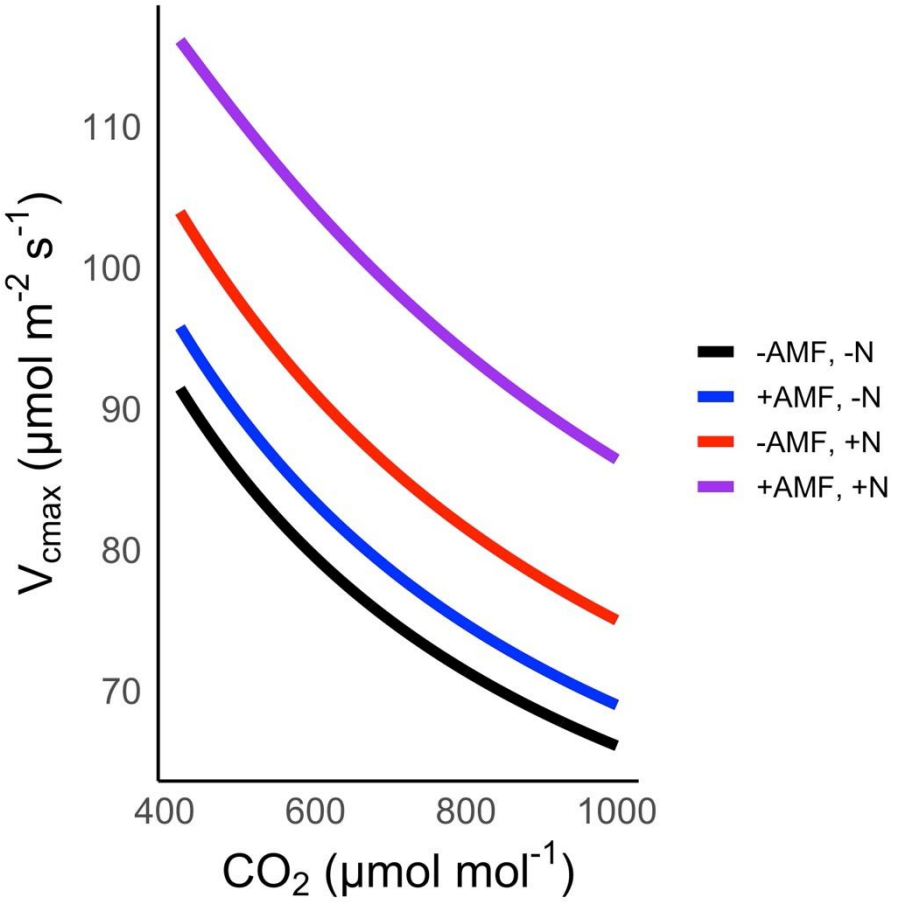
Ezekannagha and Smith (in prep)



Smith (in review)

Looking forward: mycorrhizae-CO_2 interactions

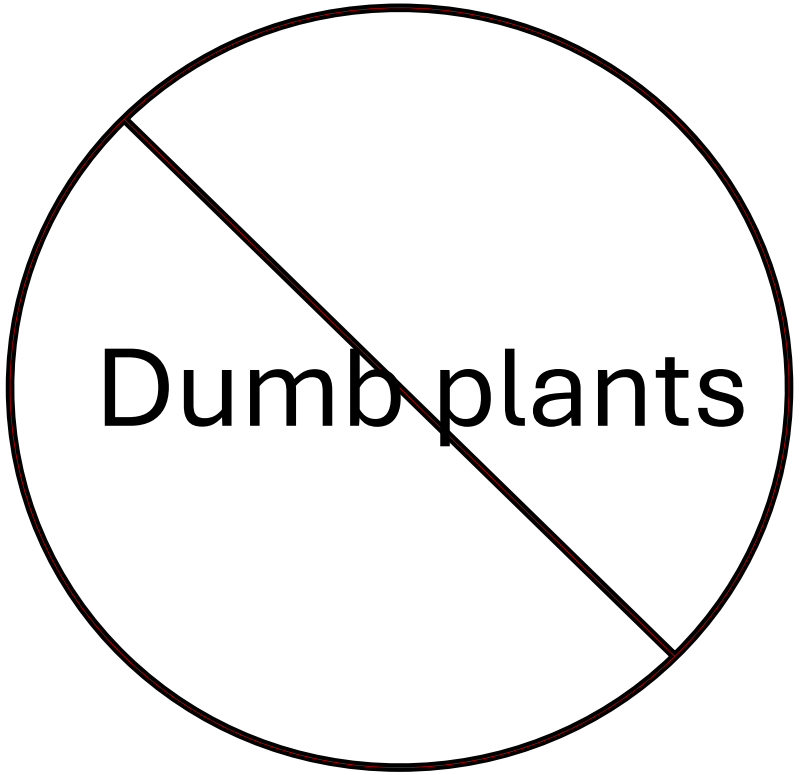




Conclusions

Conclusions

- Plants aren’t dumb!
  - Assuming plants don’t dynamically respond to their environment can lead to poor understanding of ecological functioning from individuals to the Earth system



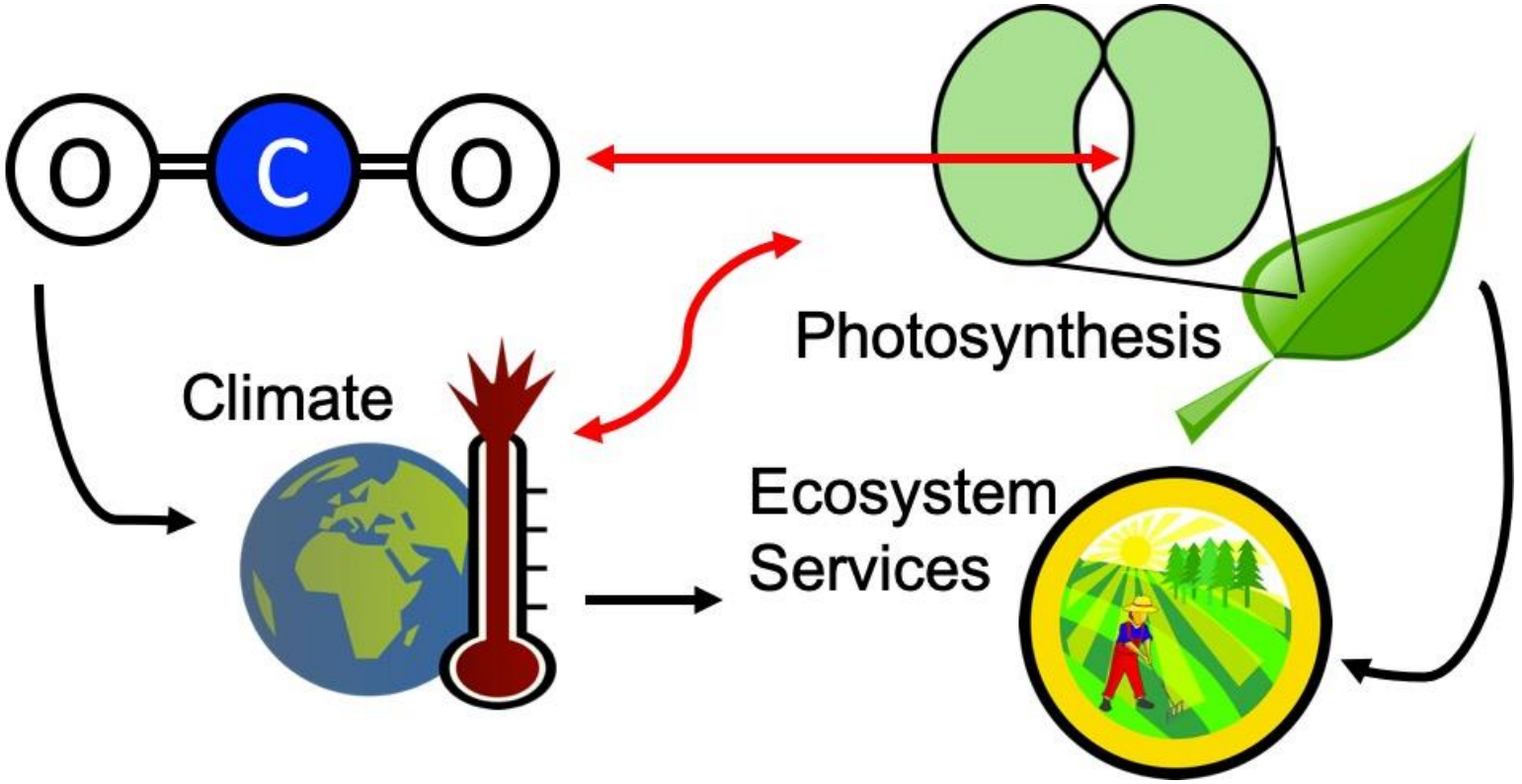
Conclusions

Quantified physiological theory can:

1. Improve mechanistic understanding of physiological processes that underlie higher level responses
2. Produce more reliable future projections



And ultimately lead to a better understanding of how future ecosystem services will be impacts by global change



Smith (2024) *Am J Botany*

Presentation available at: github.com/SmithEcophysLab/seminar/

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